2MVAb:VECTOR CALCULUS

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Contents

1. Vectors

We will mostly be working in two- and three-dimensional spaces, i.e. *plane* and *space*, respectively.

1.1. Basic properties.

A vector $\boldsymbol{A} = (a_1, a_2, a_3) \in \mathbb{R}^3$ in 3-dimensional space can be written as

$$\boldsymbol{A} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{\mathrm{k}},$$

where $\hat{\imath}, \hat{\jmath}, \hat{\mathrm{k}}$ denote the unit vectors along the coordinate directions

$$\hat{\imath} = (1, 0, 0),\ \ \hat{\jmath} = (0, 1, 0),\ \ \hat{\mathrm{k}} = (0, 0, 1).$$

The length (or magnitude) of $\boldsymbol{A}$ is a nonnegative *scalar*, denoted by $|\boldsymbol{A}|$. The direction of a vector $\boldsymbol{A}$ is sometimes denoted by $\mathrm{dir}(\boldsymbol{A})$.

Poll 1. *What is the length of the vector* $\boldsymbol{A} = (3, 2, 1)$*?*

$$(a)\, 6 \quad (b)\, \sqrt{14} \quad (c)\, 14 \quad (d)\, \sqrt{6} \quad (e)\, \textit{Something else.}$$

In general, the length of a vector $\boldsymbol{A} = (a_1, a_2, a_3)$ is given by the expression

$$|\boldsymbol{A}| = \sqrt{a_1^2 + a_2^2 + a_3^2},$$

as can be readily seen via two successive applications of Pythagoras Theorem. In particular, the answer to Poll 1 is (b).
Vectors can be added to each other: If $\boldsymbol{A} = (a_1, a_2, a_3)$ and $\boldsymbol{B} = (b_1, b_2, b_3)$, then their sum is defined as

$$\boldsymbol{A} + \boldsymbol{B} = (a_1 + b_1, a_2 + b_2, a_3 + b_3).$$

The geometry behind this operation is given by the parallelogram law: The sum $\boldsymbol{A} + \boldsymbol{B}$ is obtained by placing the vectors $\boldsymbol{A}, \boldsymbol{B}$ head to tail, and then drawing the vector from the free tail to the free head.
We can also multiply a vector by a scalar: If c is a scalar and $\boldsymbol{A}$ is a vector, then $c\boldsymbol{A}$ is the vector whose length is $|c|$ times the length of $\boldsymbol{A}$ and whose direction is the same as $\boldsymbol{A}$ if $c > 0$ and is opposite to $\boldsymbol{A}$ if $c < 0$. If $c = 0$ or $\boldsymbol{A} = \mathbf{0}$, then $c\boldsymbol{A} = \mathbf{0}$.

1.2. **Dot Product.**

Definition 2. *The* **dot product** *of the 3-dimensional vectors* $\boldsymbol{A} = (a_1, a_2, a_3)$ *and* $\boldsymbol{B} = (b_1, b_2, b_3)$ *is the scalar given by*

$$\boldsymbol{A} \cdot \boldsymbol{B} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

In the two-dimensional setting, the dot product is defined analogously:

$$(a_1, a_2) \cdot (b_1, b_2) = a_1 b_1 + a_2 b_2.$$

From the definition, it follows immediately that

$$\boldsymbol{A} \cdot \boldsymbol{B} = \boldsymbol{B} \cdot \boldsymbol{A}, \tag{1}$$

in both 2- and 3-dimensional space. We also have that

$$\boldsymbol{A} \cdot (c\boldsymbol{B} + \boldsymbol{C}) = c\boldsymbol{A} \cdot \boldsymbol{B} + \boldsymbol{A} \cdot \boldsymbol{C}, \tag{2}$$

for any scalar c. Geometrically, the dot product is given by

$$\boldsymbol{A} \cdot \boldsymbol{B} = |\boldsymbol{A}||\boldsymbol{B}| \cos\theta, \tag{3}$$

where θ is the angle between $\boldsymbol{A}$ and $\boldsymbol{B}$. In particular, the quantity $\boldsymbol{A} \cdot \boldsymbol{B}$ is positive if $\theta < \frac{\pi}{2}$, negative if $\theta > \frac{\pi}{2}$, and zero if $\theta = \frac{\pi}{2}$. We also have that

$$\boldsymbol{A} \cdot \boldsymbol{A} = |\boldsymbol{A}||\boldsymbol{A}| \underbrace{\cos 0}_{=1} = |\boldsymbol{A}|^2. \tag{4}$$

Let us check that identity (3) follows from the definition of the dot product. With that purpose in mind, consider the triangle formed by the vectors $\boldsymbol{A}$, $\boldsymbol{B}$ and $\boldsymbol{C} := \boldsymbol{A} - \boldsymbol{B}$. Then, by the Law of Cosines,

$$|\boldsymbol{C}|^2 = |\boldsymbol{A}|^2 + |\boldsymbol{B}|^2 - 2|\boldsymbol{A}||\boldsymbol{B}| \cos\theta, \tag{5}$$

where θ denotes the angle between $\boldsymbol{A}$ and $\boldsymbol{B}$ as before. On the other hand, we have that

$$\begin{aligned} |\boldsymbol{C}|^2 = \boldsymbol{C} \cdot \boldsymbol{C} &= (\boldsymbol{A} - \boldsymbol{B}) \cdot (\boldsymbol{A} - \boldsymbol{B}) \\ &= \boldsymbol{A} \cdot \boldsymbol{A} - \boldsymbol{A} \cdot \boldsymbol{B} - \boldsymbol{B} \cdot \boldsymbol{A} + \boldsymbol{B} \cdot \boldsymbol{B} \\ &= |\boldsymbol{A}|^2 + |\boldsymbol{B}|^2 - 2\boldsymbol{A} \cdot \boldsymbol{B}, \end{aligned} \tag{6}$$

where we used properties (1), (2) and (4) of the dot product. Comparing (5) and (6) yields (3), as desired.

Dot products have many useful applications, and we now turn to some of them.

1.2.1. *Computing lengths and angles.* Consider the triangle in 3-dimensional space with vertices given by

$$P = (1,0,0),\ Q = (0,1,0),\ R = (0,0,2).$$

Question 3. *What is the value of the angle $\theta := \angle(QPR)$?*

Start by noting that

$$\begin{aligned}\overrightarrow{PQ} &= Q - P = (0,1,0) - (1,0,0) = (-1,1,0),\\ \overrightarrow{PR} &= R - P = (0,0,2) - (1,0,0) = (-1,0,2).\end{aligned}$$

We have that

$$\overrightarrow{PQ} \cdot \overrightarrow{PR} = |\overrightarrow{PQ}||\overrightarrow{PR}| \cos\theta,$$

and therefore

$$\cos\theta = \frac{\overrightarrow{PQ} \cdot \overrightarrow{PR}}{|\overrightarrow{PQ}||\overrightarrow{PR}|} = \frac{(-1,1,0)\cdot(-1,0,2)}{\sqrt{(-1)^2+1^2+0^2}\sqrt{(-1)^2+0^2+2^2}} = \frac{1+0+0}{\sqrt{2}\sqrt{5}} = \frac{1}{\sqrt{10}}.$$

It follows that[1]

$$\theta = \cos^{-1}(\tfrac{1}{\sqrt{10}}) \simeq 71.5°.$$

1.2.2. *Detecting orthogonality.* Let us start with a

Poll 4. *What is the region described by the equation*

$$x + 2y + 3z = 0? \tag{7}$$

(a) Line (b) Plane (c) Point (d) Something else.

Let $P = (x,y,z)$ be a generic point in 3-dimensional space, and let $\boldsymbol{A} = (1,2,3)$. Then:

$$\overrightarrow{OP} \cdot \boldsymbol{A} = (x,y,z)\cdot(1,2,3) = x + 2y + 3z.$$

Therefore $x + 2y + 3z = 0$ if and only if $\overrightarrow{OP} \cdot \boldsymbol{A} = 0$. It follows that equation (7) defines a plane through the origin, perpendicular to $\boldsymbol{A}$. Therefore the correct answer is (b). Here we are using the fact that, given nonzero vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ determining an angle $\theta \in [0,\pi]$,

$$\boldsymbol{A}\cdot\boldsymbol{B} = 0 \Leftrightarrow \cos\theta = 0 \Leftrightarrow \theta = \frac{\pi}{2} \Leftrightarrow \boldsymbol{A} \perp \boldsymbol{B}.$$

[1] Calculator allowed for the very last computation!

1.2.3. *Finding the components of a vector* $\boldsymbol{A}$ *along a direction* $\hat{u}$. If $\hat{u}$ is a unit vector, and θ denotes the angle between $\boldsymbol{A}$ and $\hat{u}$, then the component of $\boldsymbol{A}$ along $\hat{u}$ is given by

$$|\boldsymbol{A}|\cos\theta = |\boldsymbol{A}||\hat{u}|\cos\theta = \boldsymbol{A}\cdot\hat{u}.$$

Example 5. *A simple pendulum consists of a relatively massive object hung by a string from a fixed support. It typically hangs vertically in its equilibrium position. When the pendulum is displaced from equilibrium and then released, it begins its back and forth vibration about its fixed equilibrium position. Let* $\boldsymbol{F}$ *denote the weight of the pendulum. Denote by* $\hat{T}$ *the unit tangent to the trajectory of the pendulum, and by* $\hat{\mathrm{n}}$ *the unit normal to the trajectory of the pendulum. Then the component of* $\boldsymbol{F}$ *along* $\hat{T}$ *(that is,* $\boldsymbol{F}\cdot\hat{T}$*) is what causes the pendulum to swing, and the component of* $\boldsymbol{F}$ *along* $\hat{\mathrm{n}}$ *(that is,* $\boldsymbol{F}\cdot\hat{\mathrm{n}}$*) is responsible for the tension of the string.*

1.2.4. *Areas.* So far we have discussed angles and lengths. What about areas? Can dot products help us compute, say, the area of a polygon in the plane? A polygon can be decomposed into a finite number of triangles, so let us first ask ourselves the easier question: How to compute the area of a given triangle in terms of dot products? If two sides of the triangle are given by vectors $\boldsymbol{A}$ and $\boldsymbol{B}$, and θ denotes the angle between them, then the area of the triangle equals[2]

$$\pm\frac{1}{2}|\boldsymbol{A}||\boldsymbol{B}|\sin\theta.$$

Knowing the vectors $\boldsymbol{A} = (a_1, a_2)$ and $\boldsymbol{B} = (b_1, b_2)$, we could compute $\boldsymbol{A}\cdot\boldsymbol{B}$, and then find $\cos\theta$ as in §1.2.1 above, from where $\sin\theta$ could be determined by resorting to the fundamental trigonometric identity

$$\cos^2\theta + \sin^2\theta = 1.$$

Alternatively, let $\boldsymbol{A}'$ denote the vector $\boldsymbol{A}$ rotated by $\frac{\pi}{2}$ in the counterclockwise direction, and let θ' denote the angle between $\boldsymbol{A}'$ and $\boldsymbol{B}$. Note that $\boldsymbol{A}' = (-a_2, a_1)$ since

$$\boldsymbol{A}\cdot\boldsymbol{A}' = (-a_2, a_1)\cdot(a_1, a_2) = (-a_2)a_1 + a_1a_2 = 0,$$

and the rotation is performed counterclockwise. It follows that

$$|\boldsymbol{A}||\boldsymbol{B}|\sin\theta = |\boldsymbol{A}'||\boldsymbol{B}|\cos\theta' = \boldsymbol{A}'\cdot\boldsymbol{B} = (-a_2, a_1)\cdot(b_1, b_2) = a_1b_2 - a_2b_1.$$

The latter quantity coincides with the determinant of the 2×2 matrix whose rows are $\boldsymbol{A}$ and $\boldsymbol{B}$,

$$\det(\boldsymbol{A}, \boldsymbol{B}) = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1.$$

[2]Note that $\sin\theta$ might be positive or negative depending on the value of θ, while areas are always nonnegative.

To extend this discussion to 3-dimensional space, consider three vectors

$$\boldsymbol{A} = (a_1, a_2, a_3),\ \boldsymbol{B} = (b_1, b_2, b_3),\ \boldsymbol{C} = (c_1, c_2, c_3).$$

Then we can perform an expansion along the first row of the matrix to obtain

$$\det(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}. \tag{8}$$

Geometrically, the quantity[3] $\pm \det(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C})$ corresponds to the volume of the parallelepiped determined by the vectors $\boldsymbol{A}, \boldsymbol{B}$ and $\boldsymbol{C}$. We will come back to this point in the next section.

1.3. **Cross Product.**

Definition 6. *The* **cross product** *of two vectors* $\boldsymbol{A} = (a_1, a_2, a_3)$ *and* $\boldsymbol{B} = (b_1, b_2, b_3)$ *in 3-dimensional space is the vector given by*

$$\begin{aligned} \boldsymbol{A} \times \boldsymbol{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \hat{\imath} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \hat{\jmath} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \hat{k} \\ &= (a_2 b_3 - a_3 b_2)\hat{\imath} - (a_1 b_3 - a_3 b_1)\hat{\jmath} + (a_1 b_2 - a_2 b_1)\hat{k}. \end{aligned}$$

Note that the cross product is only defined for 3-dimensional vectors. *It does not make sense* to define the cross product between vectors in 2-dimensional space!

From the definition it follows that

$$\boldsymbol{A} \times \boldsymbol{B} = -\boldsymbol{B} \times \boldsymbol{A},$$

and so, in particular, $\boldsymbol{A} \times \boldsymbol{A} = \boldsymbol{0}$, for every vector $\boldsymbol{A}$. The quantity $\boldsymbol{A} \times \boldsymbol{B}$ is a vector, and as such it has a length and a direction. What are their geometric meanings?

(a) The length $|\boldsymbol{A} \times \boldsymbol{B}|$ is the *area* of the parallelogram determined by $\boldsymbol{A}$ and $\boldsymbol{B}$;
(b) The direction $\mathrm{dir}(\boldsymbol{A} \times \boldsymbol{B})$ is perpendicular to the plane of the parallelogram defined by $\boldsymbol{A}$ and $\boldsymbol{B}$, and can be determined via the *right-hand rule*: Let your right hand point in the direction of $\boldsymbol{A}$, and let your four fingers point in the direction of $\boldsymbol{B}$; then your thumb will point in the direction of $\boldsymbol{A} \times \boldsymbol{B}$.

Example 7. $\hat{\imath} \times \hat{\jmath} = \hat{k}$. *This is clear from (a) and (b) above. Alternatively,*

$$\hat{\imath} \times \hat{\jmath} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 0\hat{\imath} - 0\hat{\jmath} + 1\hat{k} = \hat{k}.$$

Two sample applications of cross products follow.

[3] Again, the quantity on the right-hand side of (8) may be positive or negative, whereas volumes are always nonnegative.

1.3.1. *Volumes.* Let us compute the volume V of the parallelepiped determined by three given vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ in 3-dimensional space. The volume equals the area of the base times the height, but how can we express it in terms of the given vectors? From our previous discussion we know that the area of the base determined by $\boldsymbol{B}, \boldsymbol{C}$ equals $|\boldsymbol{B} \times \boldsymbol{C}|$, and that the unit vector

$$\hat{\mathrm{n}} = \frac{\boldsymbol{B} \times \boldsymbol{C}}{|\boldsymbol{B} \times \boldsymbol{C}|}$$

is perpendicular to the base. Therefore the height of the parallelepiped equals $\boldsymbol{A} \cdot \hat{\mathrm{n}}$, and the volume is given by

$$V = |\boldsymbol{B} \times \boldsymbol{C}|(\boldsymbol{A} \cdot \hat{\mathrm{n}}) = |\boldsymbol{B} \times \boldsymbol{C}| \left(\boldsymbol{A} \cdot \frac{\boldsymbol{B} \times \boldsymbol{C}}{|\boldsymbol{B} \times \boldsymbol{C}|} \right) = \boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}).$$

The latter quantity is sometimes called the *triple product* of the vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$. In general, one has that

$$\det(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}) = \boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}). \tag{9}$$

Carefully note that the order in which the vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ appear in (9) is important!

Exercise 8. *Verify identity* (9). *Hint: Recall* (8).

1.3.2. *Equation of a plane.* Given three non-collinear point P_1, P_2, P_3 in 3-dimensional space, let us find the equation of the unique plane that contains them. A generic point $P = (x, y, z)$ belongs to that plane if and only if the parallelepiped determined by the vectors $\overrightarrow{P_1P}, \overrightarrow{P_1P_2}, \overrightarrow{P_1P_3}$ has zero volume, that is,

$$\det(\overrightarrow{P_1P}, \overrightarrow{P_1P_2}, \overrightarrow{P_1P_3}) = 0. \tag{10}$$

Alternatively, let $\boldsymbol{N}$ denote some (non-zero) vector which is perpendicular to the plane determined by P_1, P_2, P_3. Then P belongs to the plane if and only if

$$\overrightarrow{P_1P} \perp \boldsymbol{N} \Leftrightarrow \overrightarrow{P_1P} \cdot \boldsymbol{N} = 0.$$

Note that a possible choice of $\boldsymbol{N}$ is given by $\boldsymbol{N} = \overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3}$. It follows that P belongs to the plane if and only if

$$\overrightarrow{P_1P} \cdot \boldsymbol{N} = \overrightarrow{P_1P} \cdot (\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3}) = 0,$$

which in light of (9) coincides with (10). So (at least up to now) everything makes sense!

2. Curves

2.1. Lines. In 3-dimensional space, a *line* can be obtained as the intersection of two planes. Alternatively, it can be thought of as the trajectory of a moving point.

Example 9. *Find a parametric equation for the line through the points*

$$Q_0 = (-1, 2, 2), \ Q_1 = (1, 3, -1). \tag{11}$$

Let $Q(t) = (x(t), y(t), z(t))$ denote the coordinates of the moving point at time t. Assuming that $Q(0) = Q_0$ and that the point moves at constant speed along the line, we have that

$$\overrightarrow{Q_0Q}(t) = t\overrightarrow{Q_0Q_1} = t(2, 1, -3),$$

and therefore

$$Q(t) = Q(0) + \overrightarrow{Q_0Q}(t) = Q_0 + t\overrightarrow{Q_0Q_1}.$$

Since $Q_0 = (-1, 2, 2)$ and $\overrightarrow{Q_0Q_1} = Q_1 - Q_0 = (1, 3, -1) - (-1, 2, 2) = (2, 1, -3)$, this can be rewritten as

$$Q(t) = (x(t), y(t), z(t)) = (-1 + 2t, 2 + t, 2 - 3t). \tag{12}$$

Poll 10. *Let us use this to determine the relative position of the points Q_0, Q_1 given in* (11) *with respect to the plane given by*

$$x + 2y + 4y = 7.$$

Which of the following is the correct option?

(*a*) *Same side* (*b*) *Opposite sides* (*c*) *Either Q_0 or Q_1 are on the plane*

(*d*) *Both Q_0 and Q_1 are on the plane* (*e*) *Cannot decide*

For $Q_0 = (-1, 2, 2)$ we have that

$$x + 2y + 4y = (-1) + 2(2) + 4(2) = 11 > 7.$$

For $Q_1 = (1, 3, -1)$ we have that

$$x + 2y + 4y = (1) + 2(3) + 4(-1) = 3 < 7.$$

Therefore Q_0 and Q_1 lie on opposite sides of the plane, and the correct option is (b). We may further recall (12) to check that $Q(t)$ belongs to the plane if and only if

$$x(t) + 2y(t) + 4z(t) = (-1 + 2t) + 2(2 + t) + 4(2 - 3t) = -8t + 11 = 7,$$

i.e. exactly when $t = \frac{1}{2}$. This was to be expected from the fact that $7 = \frac{11+3}{2}$. Therefore the point of intersection between the line and the plane in question is $Q(\frac{1}{2}) = (0, \frac{5}{2}, \frac{1}{2})$.

More generally, one can ask for the parametric equations of an arbitrary motion in the plane or in space. Let us illustrate this in the particular case of the *cycloid*.

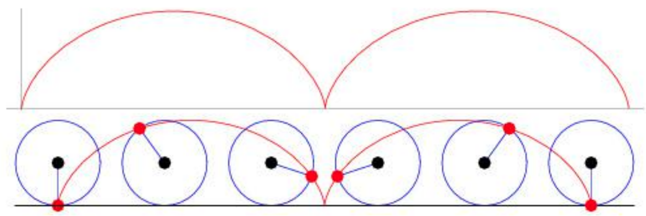

FIGURE 1. A cycloid.

2.2. **Worked out example: Cycloid.** The curve traced out by a point P on the rim of a circular wheel as the wheel rolls along a straight line without slipping is called a cycloid, see Figure 1. Assume that the wheel has radius r and rolls along the x-axis, and place the point P at the origin. Let us find parametric equations $P = (x(\theta), y(\theta))$ for the cycloid. We choose as parameter the angle of rotation θ of the wheel (so that $\theta = 0$ when P is at the origin). Let A denote the point of contact between the wheel and the x-axis and let B denote the centre of the wheel, both after rotation by θ. Elementary considerations reveal that

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BP},$$

and it turns out that these three vectors are easier to compute individually. First of all, since the rolled distance from the origin $|\overrightarrow{OA}|$ equals[4] the arc length from A to P,

$$\overrightarrow{OA} = (r\theta, 0).$$

Secondly, $\overrightarrow{AB} = (0, r)$. Finally, since $|\overrightarrow{BP}| = r$, we have that

$$\overrightarrow{BP} = (-r\sin\theta, -r\cos\theta)$$

where θ denotes the angle with the vertical direction, see Figure 2.
All in all we obtain

$$\overrightarrow{OP} = (r\theta - r\sin\theta, r - r\cos\theta),$$

or equivalently $x(\theta) = r\theta - r\sin\theta$ and $y(\theta) = r - r\cos\theta$. These are the desired parametric equations, and can be used for a number of purposes. For the time being, let us try to understand the shape of the curve near the bottom point (i.e.

[4]This is where we use the assumption that the wheel rolls *without* slipping.

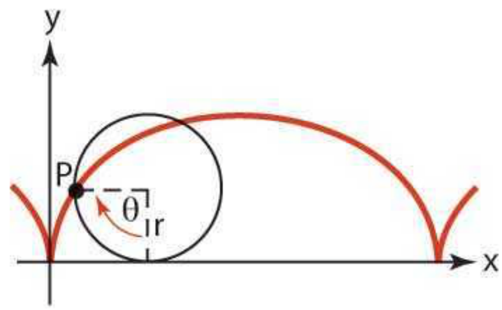

FIGURE 2. As a parameter we chose the angle of rotation θ of the wheel.

when $\theta = 0$). Assuming the radius of the wheel to be of unit length, $r = 1$, we have that

$$x(\theta) = \theta - \sin\theta, \quad y(\theta) = 1 - \cos\theta. \tag{13}$$

For sufficiently small θ, we know the approximations $\sin\theta \sim \theta$ and $\cos\theta \sim 1$, but plugging this into (13) does not yield any interesting information. To improve on this, recall Taylor's approximation

$$f(t) \sim f(0) + tf'(0) + \tfrac{t^2}{2} f''(0) + \tfrac{t^3}{6} f'''(0),$$

which is valid as long as the third derivative of f exists and is continuous at $t = 0$. In particular, for sufficiently small θ, we have that

$$\sin\theta \sim \theta - \tfrac{\theta^3}{6} \text{ and } \cos\theta \sim 1 - \tfrac{\theta^2}{2}.$$

Plugging this into (13), we see that

$$x(\theta) \sim \theta - \left(\theta - \tfrac{\theta^3}{6}\right) = \tfrac{\theta^3}{6} \text{ and } y(\theta) \sim 1 - \left(1 - \tfrac{\theta^2}{2}\right) = \tfrac{\theta^2}{2}.$$

For sufficiently small θ, we see that $|x(\theta)| \sim \frac{\theta^3}{6}$ is much smaller than $|y(\theta)| \sim \frac{\theta^2}{2}$, and that the corresponding slope is approximately equal to

$$\frac{y(\theta)}{x(\theta)} \sim \frac{\theta^2/2}{\theta^3/6} = \frac{3}{\theta},$$

which tends to ∞ as $\theta \to 0$. Therefore the slope at the origin is infinite!

2.3. **Position, velocity, acceleration.** If the *position vector* is given by

$$\boldsymbol{r}(t) = (x(t), y(t), z(t)),$$

then the *velocity* is the vector given by

$$\boldsymbol{v}(t) = \boldsymbol{r}'(t) = (x'(t), y'(t), z'(t)),$$

and the *acceleration* is the vector given by

$$\boldsymbol{a}(t) = \boldsymbol{v}'(t) = \boldsymbol{r}''(t) = (x''(t), y''(t), z''(t)).$$

The velocity vector should not be confused with *speed*, as the latter is a scalar quantity. It is given by the magnitude of the velocity vector,

$$|\boldsymbol{v}(t)| = |\boldsymbol{r}'(t)| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}.$$

Example 11. *We have seen that the position vector for the cycloid generated by a wheel of radius 1 rolling at unit speed is given by*

$$\boldsymbol{r}(t) = (t - \sin t, 1 - \cos t).$$

It follows that the velocity vector is given by $\boldsymbol{v}(t) = (1 - \cos t, \sin t)$*, and that the acceleration vector is given by* $\boldsymbol{a}(t) = (\sin t, \cos t)$*. In particular,* $\boldsymbol{v}(0) = (0, 0)$ *and* $\boldsymbol{a}(0) = (0, 1)$*. Finally, the speed is given by*

(14)
$$|\boldsymbol{v}(t)| = \sqrt{(1 - \cos t)^2 + (\sin t)^2} = \sqrt{1 - 2\cos t + \cos^2 t + \sin^2 t} = \sqrt{2 - 2\cos t}.$$

A curve given by a position vector $\boldsymbol{r}(t)$ on an interval I is called *smooth* if the velocity vector $\boldsymbol{v} = \boldsymbol{r}'$ is a continuous function of t and $\boldsymbol{v}(t) \neq \mathbf{0}$, except possibly at the endpoints of I. A curve that is made up of a finite number of smooth pieces is called *piecewise-smooth*.

2.4. **Arc length and unit tangent vector.** Suppose that C is a piecewise-smooth curve given by the position vector $\boldsymbol{r}(t) = (x(t), y(t), z(t))$, $a \leq t \leq b$, and that C is transversed exactly once as t increases from a to b. We define its *arc length function* $t \mapsto s(t)$ by

$$s(t) = \int_a^t |\boldsymbol{r}'(u)| \,\mathrm{d}u = \int_a^t \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}u}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}u}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}u}\right)^2} \,\mathrm{d}u. \tag{15}$$

The arc length $s(t)$ corresponds to the distance travelled along the trajectory from time a to t. Using the Fundamental Theorem of Calculus, we can differentiate both sides of (15) with respect to t, and obtain

$$s'(t) = |\boldsymbol{r}'(t)| = |\boldsymbol{v}(t)|. \tag{16}$$

It is often useful to parametrise a curve with respect to arc length because arc length arises naturally from the shape of the curve and does not depend on a particular coordinate system.

As an example, note that (14) and (15) together imply that the length of a complete arc of cycloid if given by

$$\int_0^{2\pi} \sqrt{2 - 2\cos u}\, \mathrm{d}u = 8.$$

The latter integral is not so easy to compute directly, but it can be done using a neat trick from multivariable calculus.

Recall that if C is a smooth curve defined by the position vector $\boldsymbol{r}(t)$, then $\boldsymbol{v}(t) = \boldsymbol{r}'(t) \neq \mathbf{0}$. The *unit tangent vector*, denoted $\hat{T}(t)$, is defined to be

$$\hat{T}(t) = \frac{\boldsymbol{v}(t)}{|\boldsymbol{v}(t)|},$$

and indicates the direction of the curve. From the Chain Rule and (16), it follows that

$$\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s}\frac{\mathrm{d}s}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s}|\boldsymbol{v}|,$$

and so $\hat{T} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s}$. This matches with what we already knew: The magnitude of the velocity vector is given by the speed $|\boldsymbol{v}| = \frac{\mathrm{d}s}{\mathrm{d}t}$, and the direction of the velocity vector is given by the unit tangent vector $\hat{T} = \frac{\boldsymbol{v}}{|\boldsymbol{v}|} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s}$. In other words, the unit tangent vector measures the rate of change of the position vector with respect to arc length.

2.5. **Kepler's Second Law.** In the early seventeenth century the German astronomer Johannes Kepler noticed that the planets revolve around the sun in an elliptical orbit with the sun at one focus, and that the line joining the sun to a planet sweeps out equal areas in equal times. Sir Isaac Newton *later* explained this using his theory of gravitational attraction. Let us try to understand Kepler's law in terms of the vector calculus which we have developed so far.

Place the sun at the origin of 3-dimensional space. Let $\boldsymbol{r}, \boldsymbol{v}$ denote the position and velocity vectors, respectively, and let $\triangle\boldsymbol{r}$ denote the displacement taking place in a certain amount of time $\triangle t$. Then the area swept in time $\triangle t$ is approximately given by

$$\text{Area} \sim \tfrac{1}{2}|\boldsymbol{r} \times \triangle\boldsymbol{r}| \sim \tfrac{1}{2}|\boldsymbol{r} \times \boldsymbol{v}|\triangle t.$$

Kepler's law then states that the quantity $|\boldsymbol{r} \times \boldsymbol{v}|$ is constant in time. Moreover, if the motion indeed occurs in a plane, then the direction of the vector $\boldsymbol{r} \times \boldsymbol{v}$ must be perpendicular to the plane of motion (since the latter contains both $\boldsymbol{r}$ and $\boldsymbol{v}$).

Therefore Kepler's law holds if and only if

$$\begin{aligned}\boldsymbol{r}\times\boldsymbol{v} \text{ is a constant vector in } t &\Leftrightarrow \frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{r}\times\boldsymbol{v}) = \mathbf{0}\\ &\Leftrightarrow \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\times\boldsymbol{v} + \boldsymbol{r}\times\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \mathbf{0}\\ &\Leftrightarrow \underbrace{\boldsymbol{v}\times\boldsymbol{v}}_{=\mathbf{0}} + \boldsymbol{r}\times\boldsymbol{a} = \mathbf{0}\\ &\Leftrightarrow \boldsymbol{r}\times\boldsymbol{a} = \mathbf{0},\end{aligned}$$

which happens if and only if the vectors $\boldsymbol{r}, \boldsymbol{a}$ are parallel. By Newton's Second Law of Motion, $\boldsymbol{F} = m\boldsymbol{a}$, this happens if and only if the gravitational force $\boldsymbol{F}$ is parallel to the position vector $\boldsymbol{r}$, which in turn is a consequence of Newton's Law of Gravitation,

$$\boldsymbol{F} = -\frac{GMm}{|\boldsymbol{r}|^3}\boldsymbol{r}. \tag{17}$$

3. Vector fields in the plane

A *vector field* in the plane is a function $\boldsymbol{F} = M\hat{\imath} + N\hat{\jmath}$, where both M and N are functions of x, y. In other words, $\boldsymbol{F} : \mathbb{R}^2 \to \mathbb{R}^2$, $(x, y) \mapsto \boldsymbol{F}(x, y)$, where for each point $(x, y) \in \mathbb{R}^2$ in the plane, $\boldsymbol{F}(x, y)$ is a vector that depends on x, y. Examples include the velocity field of a moving fluid, or a force field like a gravitational or electric field.

Example 12. $\boldsymbol{F}_1 = 2\hat{\imath} + \hat{\jmath}$ *is a constant vector field.*

Example 13. $\boldsymbol{F}_2 = x\hat{\imath}$ *does not depend on the y variable.*

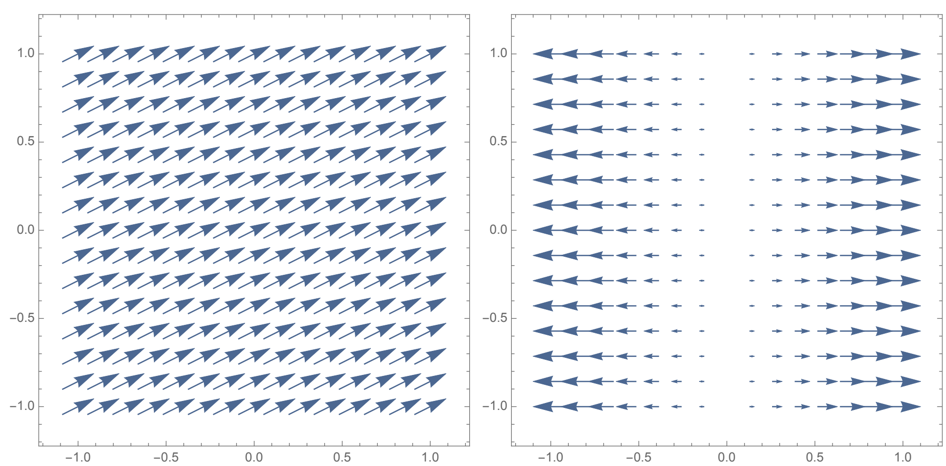

Figure 3. The vector fields $\boldsymbol{F}_1 = 2\hat{\imath} + \hat{\jmath}$ and $\boldsymbol{F}_2 = x\hat{\imath}$.

Example 14. $\boldsymbol{F}_3 = x\hat{\imath} + y\hat{\jmath}$ *points radially outwards, and its magnitude is proportional to the distance form the origin.*

Example 15. $\boldsymbol{F}_4 = -y\hat{\imath} + x\hat{\jmath}$ *corresponds to a velocity field of uniform rotation around the origin at unit angular velocity (i.e. 1 radian per time unit).*

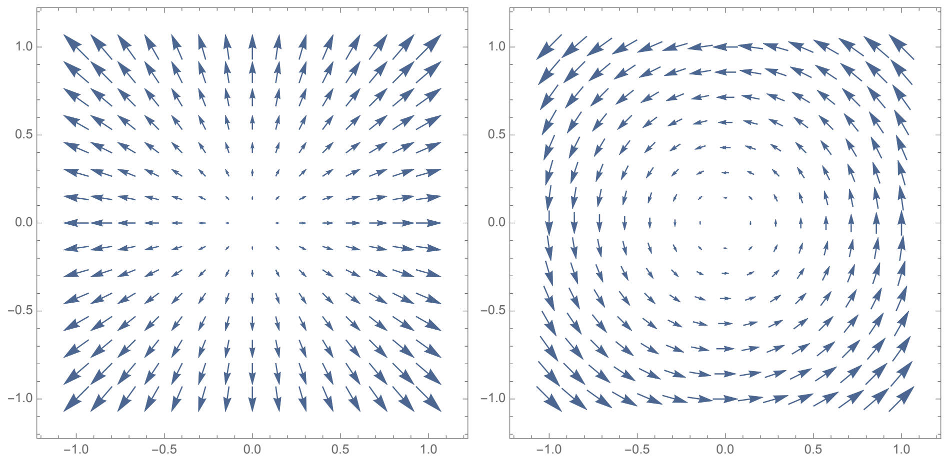

FIGURE 4. The vector fields $\boldsymbol{F}_3 = x\hat{\imath} + y\hat{\jmath}$ and $\boldsymbol{F}_4 = -y\hat{\imath} + x\hat{\jmath}$.

Drawing a given vector field can lead to useful geometric insights, but it can also sometimes be a hard task to perform without a computer.

3.1. **Work and line integrals.** From physics we know that "work equals force times distance". Along a given trajectory C parametrised on an interval $a \le t \le b$, the work done by a force field $\boldsymbol{F}$ is given by the *line integral*

$$\text{Work done by } \boldsymbol{F} \text{ along } C = \int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_a^b \boldsymbol{F} \cdot \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\,\mathrm{d}t.$$

Example 16. *Let us compute the work done by the vector field* $\boldsymbol{F} = -y\hat{\imath} + x\hat{\jmath}$ *along the trajectory* C *given by the arc of parabola*

$$C : t \mapsto (x(t), y(t)) = (t, t^2), \quad 0 \le t \le 1. \tag{18}$$

Along C*, we have that* $\frac{\mathrm{d}x}{\mathrm{d}t} = 1$ *and* $\frac{\mathrm{d}y}{\mathrm{d}t} = 2t$ *and moreover* $\boldsymbol{F} = (-y, x) = (-t^2, t)$*. It follows that*

$$\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_C \boldsymbol{F} \cdot \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\,\mathrm{d}t = \int_0^1 (-t^2, t) \cdot (1, 2t)\,\mathrm{d}t = \int_0^1 t^2\,\mathrm{d}t = \tfrac{1}{3}.$$

Alternatively, we have that $\boldsymbol{F} = (M, N)$ *where* $M(x, y) = -y$ *and* $N(x, y) = x$*, and* $\mathrm{d}\boldsymbol{r} = (\mathrm{d}x, \mathrm{d}y)$*. Therefore* $\boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = M\,\mathrm{d}x + N\,\mathrm{d}y$*, and so*

$$\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_C M\,\mathrm{d}x + N\,\mathrm{d}y.$$

To evaluate the latter integral, we express x, y *in terms of a single variable* t*, and then substitute to get*

$$\int_C \boldsymbol{F} \cdot\, \mathrm{d}\boldsymbol{r} = \int_C -y\,\mathrm{d}x + x\,\mathrm{d}y = \int_C -t^2\,\mathrm{d}t + t(2t\,\mathrm{d}t) = \int_0^1 t^2\,\mathrm{d}t = \tfrac{1}{3},$$

where we again used that $x = t$ *and* $y = t^2$ *and therefore* $\mathrm{d}x = \mathrm{d}t$ *and* $\mathrm{d}y = 2t\,\mathrm{d}t$*.*

Carefully note that the line integral $\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}$ may depend on the trajectory C, but *not* on the parametrisation used. For instance, instead of (18) we could have used

$$C : \theta \mapsto (x(\theta), y(\theta)) = (\sin\theta, \sin^2\theta), \;\; 0 \le \theta \le \tfrac{\pi}{2}.$$

This would lead to more complicated integrals but ultimately yield the same result $(\frac{1}{3})$.

We now present a more *geometric* approach to computing line integrals, which sometimes can lead to significant simplifications. From §2.4, we know that

$$\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \left(\frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}\right) = \hat{T}\,\frac{\mathrm{d}s}{\mathrm{d}t},$$

where $\hat{T}$ denotes the unit tangent vector, and $\frac{\mathrm{d}s}{\mathrm{d}t}$ corresponds to the speed. Therefore $\mathrm{d}\boldsymbol{r} = \hat{T}\,\mathrm{d}s$, and

$$\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_C \boldsymbol{F} \cdot \hat{T}\,\mathrm{d}s. \tag{19}$$

Example 17. *Let us compute the work done by the force field* $\boldsymbol{F} = x\hat{\imath} + y\hat{\jmath}$ *along a circle* C_a *of radius* $a > 0$ *centred at the origin (transversed counterclockwise).*

If $\hat{T}$ denotes the unit vector tangent to the circle C_a, then $\boldsymbol{F} \perp \hat{T}$, and so $\boldsymbol{F} \cdot \hat{T} = 0$. Consequently,

$$\int_{C_a} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_{C_a} \boldsymbol{F} \cdot \hat{T}\,\mathrm{d}s = 0.$$

Example 18. *Let us now compute the work done by the force field* $\boldsymbol{F} = -y\hat{\imath} + x\hat{\jmath}$ *along the same circle* C_a *of radius* $a > 0$ *centered at the origin (transversed counterclockwise).*

In this case, $\boldsymbol{F} \parallel \hat{T}$ and so

$$\boldsymbol{F} \cdot \hat{T} = |\boldsymbol{F}| = \sqrt{(-y)^2 + x^2} = \sqrt{x^2 + y^2} = a,$$

for every $(x, y) \in C_a$. Consequently,

$$\int_{C_a} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_{C_a} \boldsymbol{F} \cdot \hat{T} \,\mathrm{d}s = a \int_{C_a} \mathrm{d}s = a \,\mathrm{length}(C_a) = a(2\pi a) = 2\pi a^2.$$

This is shorter than any of the previous methods, which would involve e.g. parametrising the circle

$$C_a : \theta \mapsto (x(\theta), y(\theta)) = (a\cos\theta, a\sin\theta), \ \ 0 \leq \theta \leq 2\pi,$$

and computing the integral

$$\begin{aligned}\int_{C_a} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_{C_a} -y \,\mathrm{d}x + x \,\mathrm{d}y &= \int_{C_a} -(a\sin\theta)(-a\sin\theta \,\mathrm{d}\theta) + (a\cos\theta)(a\cos\theta \,\mathrm{d}\theta) \\ &= \int_0^{2\pi} a^2 (\underbrace{\sin^2\theta + \cos^2\theta}_{=1}) \,\mathrm{d}\theta = 2\pi a^2.\end{aligned}$$

3.2. **The fundamental theorem of calculus for line integrals.** We start with a concrete example. Consider the vector field $\boldsymbol{F} = y\hat{\imath} + x\hat{\jmath}$, and the closed path $C = C_1 + C_2 + C_3$ enclosing the sector of the unit disc corresponding to $0 \leq \theta \leq \frac{\pi}{4}$, transversed counterclockwise from the origin. Let us compute the line integral

$$\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_C y \,\mathrm{d}x + x \,\mathrm{d}y. \tag{20}$$

Along C_1: From $(0, 0)$ to $(1, 0)$ along the x-axis. Since $y = 0$ on C_1, we have that $\mathrm{d}y = 0$, and so

$$\int_{C_1} y \,\mathrm{d}x + x \,\mathrm{d}y = \int_{C_1} 0 \,\mathrm{d}x + 0 = 0. \tag{21}$$

This can also be seen geometrically as follows: along the x-axis, the vector field $\boldsymbol{F}$ is parallel to the vector $\hat{\jmath}$, whereas the unit tangent vector $\hat{T}$ points in the direction of $\hat{\imath}$, and so $\boldsymbol{F} \cdot \hat{T} = 0$. From (19) it then follows that the work done by $\boldsymbol{F}$ along C_1 is zero, in accordance with (21).

Along C_2: The portion of the unit circle corresponding to C_2 can be parametrised as follows:

$$C_2 : \theta \mapsto (x(\theta), y(\theta)) = (\cos\theta, \sin\theta), \ \ 0 \leq \theta \leq \tfrac{\pi}{4}.$$

Since $\mathrm{d}x = -\sin\theta\,\mathrm{d}\theta$ and $\mathrm{d}y = \cos\theta\,\mathrm{d}\theta$, we then have that

$$\begin{aligned}\int_{C_2} y\,\mathrm{d}x + x\,\mathrm{d}y &= \int_0^{\frac{\pi}{4}} (\sin\theta)(-\sin\theta\,\mathrm{d}\theta) + (\cos\theta)(\cos\theta\,\mathrm{d}\theta) \\ &= \int_0^{\frac{\pi}{4}} (\underbrace{\cos^2\theta - \sin^2\theta}_{=\cos 2\theta})\,\mathrm{d}\theta = \left[\frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{4}} = \tfrac{1}{2}. \qquad (22)\end{aligned}$$

Along C_3: The path C_3 corresponds to the line segment from $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ to $(0,0)$, which can be parametrised by

$$C_3 : t \mapsto \left(\tfrac{1-t}{\sqrt{2}}, \tfrac{1-t}{\sqrt{2}}\right), \quad 0 \leq t \leq 1.$$

To simplify the computations ahead, we can instead parametrise the *inverse path* $-C_3$ (i.e. C_3 transversed backwards) by

$$-C_3 : t \mapsto (t,t), \quad 0 \leq t \leq \tfrac{1}{\sqrt{2}},$$

and then note that $\int_{-C_3} = -\int_{C_3}$. We have that

$$\int_{-C_3} y\,\mathrm{d}x + x\,\mathrm{d}y = \int_0^{\frac{1}{\sqrt{2}}} t\,\mathrm{d}t + t\,\mathrm{d}t = \int_0^{\frac{1}{\sqrt{2}}} 2t\,\mathrm{d}t = \left[t^2\right]_0^{\frac{1}{\sqrt{2}}} = \tfrac{1}{2},$$

and therefore

$$\int_{C_3} y\,\mathrm{d}x + x\,\mathrm{d}y = -\tfrac{1}{2}.$$

All in all, we conclude that the total work equals

$$\int_C = \int_{C_1} + \int_{C_2} + \int_{C_3} = 0 + \tfrac{1}{2} - \tfrac{1}{2} = 0,$$

which means that we did a lot of computations for (literally) nothing!

Let us now try to understand how to predict the value of some line integrals and avoid such unnecessarily lengthy computations. Assume that the vector field $\boldsymbol{F}$ satisfies

$$\boldsymbol{F} = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right),$$

for some function $f : \mathbb{R}^2 \to \mathbb{R}$. In other words, $\boldsymbol{F}$ is a *gradient field* (and f is sometimes called the *potential* of $\boldsymbol{F}$). In this case, the evaluation of the line integral $\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}$ can be simplified with the help of the following important result.

Theorem 19 (Fundamental Theorem of Calculus for Line Integrals)**.** *Let C be a smooth curve given by the position vector $\boldsymbol{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function whose gradient ∇f is continuous on C. Then:*

$$\int_C \nabla f \cdot \mathrm{d}\boldsymbol{r} = f(\boldsymbol{r}(b)) - f(\boldsymbol{r}(a)).$$

Writing $\mathrm{d}\boldsymbol{r} = (\mathrm{d}x, \mathrm{d}y)$, the gradient $\nabla f = (f_x, f_y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$, and

$$\mathrm{d}f = f_x \, \mathrm{d}x + f_y \, \mathrm{d}y = \nabla f \cdot \mathrm{d}\boldsymbol{r}$$

for the differential of f, we see that

$$\int_C \nabla f \cdot \mathrm{d}\boldsymbol{r} = \int_C \mathrm{d}f = f(\boldsymbol{r}(b)) - f(\boldsymbol{r}(a)).$$

Thus Theorem 19 can be seen as an extension of the Fundamental Theorem of Calculus from single-variable calculus. A rigorous proof follows in the two-dimensional case when $C \subset \mathbb{R}^2$ and $f : \mathbb{R}^2 \to \mathbb{R}$. The reader is encouraged to check that the same proof applies when $C \subset \mathbb{R}^3$ is a curve in three-dimensional space (and accordingly $f : \mathbb{R}^3 \to \mathbb{R}$).

Proof of Theorem 19. We have that

$$\int_C \nabla f \cdot \mathrm{d}\boldsymbol{r} = \int_C f_x \, \mathrm{d}x + f_y \, \mathrm{d}y = \int_C \left(f_x \frac{\mathrm{d}x}{\mathrm{d}t} + f_y \frac{\mathrm{d}y}{\mathrm{d}t} \right) \mathrm{d}t.$$

On the other hand, parametrising

$$C : t \mapsto \boldsymbol{r}(t) = (x(t), y(t)), \quad a \leq t \leq b,$$

leads to $\mathrm{d}x = x'(t) \, \mathrm{d}t$ and $\mathrm{d}y = y'(t) \, \mathrm{d}t$. We then have by the Chain Rule that

$$\begin{aligned} \int_C \nabla f \cdot \mathrm{d}\boldsymbol{r} = \int_C \left(f_x \frac{\mathrm{d}x}{\mathrm{d}t} + f_y \frac{\mathrm{d}y}{\mathrm{d}t} \right) \mathrm{d}t &= \int_a^b \frac{\mathrm{d}f}{\mathrm{d}t}(x(t), y(t)) \mathrm{d}t \\ &= [f(x(t), y(t))]_a^b = f(\boldsymbol{r}(b)) - f(\boldsymbol{r}(a)). \end{aligned}$$

Here we used the Fundamental Theorem of Calculus from single-variable calculus to go from the first to the second line. $\square$

Example 20. *Going back to the example at the beginning of this section, we see that the vector field $\boldsymbol{F} = y\hat{\imath} + x\hat{\jmath}$ is a gradient field since $\boldsymbol{F} = \nabla f$ if $f(x, y) = xy$. In particular, we can easily compute*

$$\int_{C_2} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = f(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}) - f(0, 0) = (\tfrac{1}{\sqrt{2}})^2 - 0 = \tfrac{1}{2},$$

in accordance with (22)*. In a similar way, the fact that*

$$\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = 0$$

follows immediately from Theorem 19 since $\boldsymbol{F}$ *is a gradient field and the curve* C *is closed.*

3.3. **Consequences.** If $\boldsymbol{F} = \nabla f$ is a gradient field, then:

- The integral $\int_C F \cdot \mathrm{d}\boldsymbol{r}$ is *independent of path*: This means that if two paths C_1, C_2 have the same initial and terminal points, then
$$\int_{C_1} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_{C_2} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}.$$
- The vector field $\boldsymbol{F}$ is *conservative*: This means that[5]
$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = 0,$$
for any *closed* curve C. Indeed, by Theorem 19, we have that
$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \oint_C \nabla f \cdot \mathrm{d}\boldsymbol{r} = f(\text{end}) - f(\text{start}) = 0,$$
since the initial and terminal points of C coincide.

Example 21. *Consider the vector field* $\boldsymbol{F} = -y\hat{\imath} + x\hat{\jmath}$*, and let* C *denote the unit circle (transversed counterclockwise) with unit tangent vector* $\hat{T}$*. On the unit circle,* $\boldsymbol{F} \parallel \hat{T}$*, and so* $\boldsymbol{F} \cdot \hat{T} = |\boldsymbol{F}| = 1$*. Thus*

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \oint_C \boldsymbol{F} \cdot \hat{T}\, \mathrm{d}s = \oint_C 1\, \mathrm{d}s = \textit{length}(C) = 2\pi \neq 0.$$

In particular, we see that the vector field $\boldsymbol{F} = -y\hat{\imath} + x\hat{\jmath}$ *is not a gradient field. In other words, there exists no function* $f : \mathbb{R}^2 \to \mathbb{R}$ *such that* $\boldsymbol{F} = \nabla f$*. From the previous discussion, it follows that* $\boldsymbol{F}$ *is* not *a conservative vector field and that the line integral* $\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}$ *is* not *path-independent.*

Physical interpretation. If a force $\boldsymbol{F}$ is known to equal the gradient of a potential f, i.e. $\boldsymbol{F} = \nabla f$, then as guaranteed by Theorem 19 the work done by $\boldsymbol{F}$ along a given path equals the change of potential along that path. Examples include the gravitational and electric fields.[6] Conservativeness means that no energy can be extracted from the field "for free", which in turn translates into a Law of Conservation of Energy for the total energy of the system.

[5]The notation $\oint$ emphasises the fact that the curve C is *closed*.

[6]By contrast, it turns out that magnetic field do *not* come from a potential.

How to test whether a vector field $\boldsymbol{F} = (M, N)$ is a gradient field? A *necessary condition* is as follows: If $\boldsymbol{F} = \nabla f$, then $M = f_x$ and $N = f_y$. If f has continuous partial derivatives of second order,[7] then Clairaut's Theorem applies and ensures that the mixed second derivatives coincide, $f_{xy} = f_{yx}$, which in particular implies that

$$N_x = M_y.$$

In the converse direction, we will see in §4.1 below that if $\boldsymbol{F}$ is defined and differentiable *everywhere*, and moreover $N_x = M_y$ holds everywhere, then there exists a function $f : \mathbb{R}^2 \to \mathbb{R}$ such that $\boldsymbol{F} = \nabla f$ (i.e. $\boldsymbol{F}$ is a gradient field).

Example 22. *Consider the vector field $\boldsymbol{F} = -y\hat{\imath} + x\hat{\jmath}$ (see Figure 4). In this case, $M(x, y) = -y$ and $N(x, y) = x$, and therefore*

$$N_x = 1 \neq -1 = M_y.$$

Therefore $\boldsymbol{F}$ is not a gradient field.

Example 23. *For which values of a is the vector field*

$$\boldsymbol{F} = (4x^2 + axy)\hat{\imath} + (4x^2 + 3y^2)\hat{\jmath}$$

a gradient field? In this case, $M(x, y) = 4x^2 + axy$ and $N(x, y) = 4x^2 + 3y^2$, and so $N_x = 8x$ and $M_y = ax$. These two quantities coincide only if $a = 8$. Since $\boldsymbol{F}$ is defined everywhere in $\mathbb{R}^2$ (for any value of a), it follows that for $a = 8$ the vector field $\boldsymbol{F}$ is indeed a gradient field.

Given a vector field $\boldsymbol{F} = (M, N)$ defined everywhere in $\mathbb{R}^2$ and satisfying $N_x = M_y$, we have just seen that there exists a function $f : \mathbb{R}^2 \to \mathbb{R}$ such that $\boldsymbol{F} = \nabla f$. But how can we find such an f? There are (at least) two distinct methods, which we illustrate for the case $a = 8$ of Example 23:

Method 1. *Computing line integrals.* Let

$$\boldsymbol{F} = (4x^2 + 8xy)\hat{\imath} + (4x^2 + 3y^2)\hat{\jmath}.$$

[7] Otherwise the conclusion may fail, as illustrated in the tutorial sheet from Week 3: If $f : \mathbb{R}^2 \to \mathbb{R}$ is defined by $f(0, 0) = 0$ and $f(x, y) = \frac{xy(x^2-y^2)}{x^2+y^2}$ if $(x, y) \neq (0, 0)$, then one can (and should) check that

$$\frac{\partial^2 f}{\partial x \partial y}(0, 0) \neq \frac{\partial^2 f}{\partial y \partial x}(0, 0).$$

Take any path C connecting the origin $(0,0)$ to an arbitrary point $(x_1, y_1) \in \mathbb{R}^2$. Since $\boldsymbol{F} = \nabla f$ is a gradient field, we can apply Theorem 19. It follows that

$$\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_C \nabla f \cdot \mathrm{d}\boldsymbol{r} = f(x_1, y_1) - f(0,0),$$

or equivalently

$$f(x_1, y_1) = \int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} + f(0,0).$$

Now we make a specific choice of path C: Let $C = C_1 + C_2$, where C_1 is the horizontal line segment connecting $(0,0)$ to $(x_1, 0)$, and C_2 is the vertical line segment connecting $(x_1, 0)$ to (x_1, y_1). Then

$$\int_{C_1} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_0^{x_1} 4x^2 \, \mathrm{d}x = \tfrac{4}{3} x_1^3,$$

since along the x-axis we have that $y = 0 = \mathrm{d}y$. Similarly,

$$\int_{C_2} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_0^{y_1} (4x_1^2 + 3y^2) \, \mathrm{d}y = \left[4x_1^2 y + y^3\right]_0^{y_1} = 4x_1^2 y_1 + y_1^3,$$

since along the vertical line $x = x_1$ we have that $\mathrm{d}x = 0$. It follows that a suitable potential $f : \mathbb{R}^2 \to \mathbb{R}$ is given by

$$f(x,y) = \tfrac{4}{3} x^3 + 4x^2 y + y^3.$$

Any other admissible potential differs from this one by an additive constant, and so the general solution is given by

$$f(x,y) = \tfrac{4}{3} x^3 + 4x^2 y + y^3 + C, \tag{23}$$

for some $C \in \mathbb{R}$.

Method 2. *Computing anti-derivatives.* We aim to solve the system of differential equations

$$\begin{cases} f_x &= 4x^2 + 8xy \\ f_y &= 4x^2 + 3y^2 \end{cases} \tag{24}$$

Integrating the first equation from (24) with respect to x, we get that

$$f(x,y) = \tfrac{4}{3} x^3 + 4x^2 y + g(y), \tag{25}$$

for some function g which depends on y but not on x, and which we still have to determine. With this purpose in mind, differentiate (25) with respect to y to get

$$f_y = 4x^2 + g'(y).$$

This can be matched with the second equation from (24) to yield $g'(y) = 3y^2$, whence $g(y) = y^3 + C$. Inserting the latter into (25), we finally obtain that

$$f(x,y) = \tfrac{4}{3} x^3 + 4x^2 y + y^3 + C,$$

in accordance with (23).

3.4. **Curl.** We introduce the notion of *curl* of a vector field, which is a measure of how far a given vector field is from being conservative.

Definition 24. *Let* $\boldsymbol{F} = M\hat{\imath} + N\hat{\imath}$ *be a vector field on the plane. Its* **curl** *is given by*

$$\operatorname{curl}(\boldsymbol{F}) := N_x - M_y.$$

From our discussion in §3.3 it follows that $\operatorname{curl}(\boldsymbol{F}) = 0$ if $\boldsymbol{F}$ is conservative. The curl of a vector field has actual physical meaning. For instance, the curl of a velocity field measures the rotation component of the motion – this is sometimes referred to as the *vorticity*.

Example 25. *If* $\boldsymbol{F} = (a, b)$ *is a constant vector field, then* $\operatorname{curl}(\boldsymbol{F}) = 0$*.*

Example 26. *If* $\boldsymbol{F} = (x, y)$*, then* $\operatorname{curl}(\boldsymbol{F}) = \frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x) = 0$*. This agrees with the (geometrically obvious) fact that no rotation is taking place.*

Example 27. *If* $\boldsymbol{F} = (-y, x)$*, then* $\operatorname{curl}(\boldsymbol{F}) = \frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y) = 2 \neq 0$*.*

Example 27 shows that the curl of a velocity field measures (twice) the angular velocity at any given point. In the case of a force field, the curl measures the *torque*[8] exerted on a test object in the field.

4. Green's Theorem

4.1. **Statement.** Green's Theorem gives the relationship between a line integral around a closed curve C and a double integral over the plane region R bounded by C. It is named after the British mathematical physicist George Green, though its first proof is due to Bernhard Riemann. It is the two-dimensional special case of the more general Stokes' Theorem which we will study later on in the course.

Theorem 28 (Green's Theorem)**.** *Let* C *be a closed curve, oriented counterclockwise, and enclosing a region* $R \subset \mathbb{R}^2$*. If* $\boldsymbol{F}$ *is a vector field defined and differentiable everywhere in* R*, then*

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \iint_R \operatorname{curl}(\boldsymbol{F}) \, \mathrm{d}A. \tag{26}$$

In coordinates $\boldsymbol{F} = (M, N)$, identity (26) reads as follows:

$$\oint_C M \, \mathrm{d}x + N \, \mathrm{d}y = \iint_R (N_x - M_y) \, \mathrm{d}A.$$

[8]Recall from Physics that, similarly to Newton's Second Law $\boldsymbol{F} = \frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v})$, we have that $\boldsymbol{\tau} = \frac{\mathrm{d}}{\mathrm{d}t}(I\boldsymbol{\omega})$, where $\boldsymbol{\tau}$, I and $\boldsymbol{\omega}$ denote the torque, moment of inertia and angular velocity, respectively.

At first the statement of Green's Theorem may seem a bit strange. After all, the usual methods to compute the left- and right-hand sides of (26) are completely different! As we will see, Green's Theorem turns out to be a very powerful tool.

Example 29. *Let C be the circle of radius 1 centred at the point $(2,0)$, oriented counterclockwise. Let us compute the line integral*

$$\oint_C ye^{-x}\,\mathrm{d}x + \left(\tfrac{x^2}{2} - e^{-x}\right)\,\mathrm{d}y.$$

Doing it directly seems quite hard.[9] *Let us try to use Green's Theorem instead. Letting $\boldsymbol{F} = (M,N)$ with $M(x,y) = ye^{-x}$ and $N(x,y) = \frac{x^2}{2} - e^{-x}$, we have that*

$$\operatorname{curl}(\boldsymbol{F}) = N_x - M_y = (x + e^{-x}) - e^{-x} = x,$$

and so by Green's Theorem

$$\oint_C \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{r} = \iint_R \operatorname{curl}(\boldsymbol{F})\,\mathrm{d}A = \iint_R x\,\mathrm{d}A,$$

where R denotes the disc of radius 1 centred at $(2,0)$ enclosed by C. This integral can be computed directly (Exercise: do it). However, we can also note that

$$\iint_R x\,\mathrm{d}A = \underbrace{\operatorname{area}(R)}_{=\pi}\,\bar{x},$$

where $\bar{x}$ denotes the x-coordinate of the centre of mass of R. By symmetry, we immediately see that $\bar{x} = 2$, and so we finally conclude that

$$\oint_C ye^{-x}\,\mathrm{d}x + \left(\tfrac{x^2}{2} - e^{-x}\right)\,\mathrm{d}y = 2\pi.$$

Let us now consider a closed curve C (oriented counterclockwise) enclosing a region $R \subset \mathbb{R}^2$, and a vector field $\boldsymbol{F}$ with satisfies $\operatorname{curl}(\boldsymbol{F}) \equiv 0$ everywhere on R. By Green's Theorem, we conclude at once that

$$\oint_C \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{r} = \iint_R \operatorname{curl}(\boldsymbol{F})\,\mathrm{d}A = \iint_R 0\,\mathrm{d}A = 0.$$

In particular, this shows that if $\boldsymbol{F}$ is defined everywhere in the plane and $\operatorname{curl}(\boldsymbol{F}) \equiv 0$, then $\boldsymbol{F}$ is conservative. This was a claim made in §3.3 which we could now easily prove appealing to Green's Theorem.

[9] You are still welcome to give it a try!

4.2. **Proof.** In this section we prove Green's Theorem.

Proof of Green's Theorem. Let $\boldsymbol{F} = (M, N)$. We proceed in three steps.
Step 1. It will be enough to establish the special case when $N = 0$:

$$\oint_C M\,\mathrm{d}x = \iint_R -M_y\,\mathrm{d}A. \tag{27}$$

In fact, if we show (27), then a similar reasoning[10] yields

$$\oint_C N\,\mathrm{d}y = \iint_R N_x\,\mathrm{d}A, \tag{28}$$

and Green's Theorem will then follow from adding (27) and (28).
Step 2. We may cut the region R into *vertically simple regions* of the form

$$\{(x, y) \in \mathbb{R}^2 : \ a \le x \le b, \ \ f_1(x) \le y \le f_2(x)\}, \tag{29}$$

and it will be enough to show Green's Theorem for one such region. In fact, if we decompose $R = R_1 \cup R_2$ as the union of two smaller regions R_1, R_2 with boundaries C_1, C_2, respectively, and assume that Green's Theorem holds in the sub-regions R_1, R_2, then

$$\oint_C M\,\mathrm{d}x = \oint_{C_1} M\,\mathrm{d}x + \oint_{C_2} M\,\mathrm{d}x = \iint_{R_1} -M_y\,\mathrm{d}A + \iint_{R_2} -M_y\,\mathrm{d}A = \iint_R -M_y\,\mathrm{d}A.$$

Here the first identity holds because the boundary between R_1 and R_2 is transversed twice *with opposite orientations.*
Step 3. (Main step) Let us prove that, if R is a vertically simple region in the sense of (29), and if C is the boundary of R transversed counterclockwise, then

$$\oint_C M\,\mathrm{d}x = \iint_R -M_y\,\mathrm{d}A. \tag{30}$$

The strategy is to decompose $C = C_1 + C_2 + C_3 + C_4$ (described below), and analyse each of the line integrals that arise. First of all, since

$$C_1 : x \mapsto (x, f_1(x)), \ \ a \le x \le b,$$

we have that

$$\int_{C_1} M\,\mathrm{d}x = \int_a^b M(x, f_1(x))\,\mathrm{d}x.$$

In a similar way, since

$$C_3 : x \mapsto (x, f_2(x)), \ \ b \le x \le a,$$

[10] Exercise: Convince yourself that this is indeed the case.

we have that

$$\int_{C_3} M\,\mathrm{d}x = \int_b^a M(x, f_2(x))\,\mathrm{d}x = -\int_a^b M(x, f_2(x))\,\mathrm{d}x.$$

Finally, C_2 and C_4 correspond to vertical line segments, along which one necessarily has $\mathrm{d}x = 0$. Therefore

$$\int_{C_2} M\,\mathrm{d}x = \int_{C_4} M\,\mathrm{d}x = 0.$$

All in all, we have that

$$\oint_C M\,\mathrm{d}x = \oint_{C_1} M\,\mathrm{d}x + 0 + \oint_{C_3} M\,\mathrm{d}x + 0 = \int_a^b (M(x, f_1(x)) - M(x, f_2(x)))\,\mathrm{d}x. \tag{31}$$

On the other hand, the Fundamental Theorem of Calculus implies that

$$\iint_R -M_y\,\mathrm{d}A = -\int_a^b \left(\int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y}\,\mathrm{d}y \right) \mathrm{d}x = -\int_a^b (M(x, f_2(x)) - M(x, f_1(x))\,\mathrm{d}x,$$

which coincides with the right-hand side of (31). Therefore identity (30) holds, and the proof is complete. □

4.3. **Applications.** Since the area of a given region $R \subset \mathbb{R}^2$ equals the double integral $\iint_R 1\,\mathrm{d}A$, we wish to choose a vector field $\boldsymbol{F} = (M, N)$ so that

$$\mathrm{curl}(\boldsymbol{F}) = N_x - M_y = 1.$$

There are many possibilities, among which we highlight the following:

$$\begin{aligned} M(x,y) &= 0 \text{ and } N(x,y) = x, \\ M(x,y) &= -y \text{ and } N(x,y) = 0, \\ M(x,y) &= -\tfrac{y}{2} \text{ and } N(x,y) = \tfrac{x}{2}. \end{aligned}$$

Green's Theorem tells us that the area of R (enclosed by C) is then given by:

$$\mathrm{area}(R) = \oint_C x\,\mathrm{d}y = -\oint_C y\,\mathrm{d}x = \tfrac{1}{2}\oint_C x\,\mathrm{d}y - y\,\mathrm{d}x. \tag{32}$$

Example 30. *Recall our discussion of the cycloid from §2.2. Let us find the area under one arc of the cycloid*

$$(x(t), y(t)) = (t - \sin t, 1 - \cos t).$$

Let C_1 be the arc of cycloid from $(0,0)$ to $(2\pi,0)$, which corresponds to $0 \le t \le 2\pi$, and let C_2 be the line segment from $(0,2\pi)$ to $(0,0)$. The line segment C_2 can be parametrised by

$$C_2 : (x(t), y(t)) = (2\pi - t, 0), \quad 0 \le t \le 2\pi.$$

Then $C = C_1 + C_2$ is transversed clockwise, so $-C$ is oriented positively and encloses the region R under one arc of the cycloid. From (32) it follows that

$$\begin{aligned}
\operatorname{area}(R) &= \oint_{-C} -y\,\mathrm{d}x = \int_{C_1} y\,\mathrm{d}x + \int_{C_2} y\,\mathrm{d}x \\
&= \int_0^{2\pi} (1-\cos t)(1-\cos t)\,\mathrm{d}t + \int_0^{2\pi} 0\,(-\mathrm{d}t) \\
&= \int_0^{2\pi} (1 - 2\cos t + \cos^2 t)\,\mathrm{d}t = \int_0^{2\pi} (1 - 2\cos t + \tfrac{1+\cos(2t)}{2})\,\mathrm{d}t \\
&= \big[t - 2\sin t + \tfrac{t}{2} + \tfrac{1}{4}\sin 2t\big]_0^{2\pi} = 3\pi.
\end{aligned}$$

We reach the non-trivial conclusion that *the area under one arc of the cycloid equals three times the area of the generating circle.*

4.4. **Flux.** Let $\boldsymbol{F} : \mathbb{R}^2 \to \mathbb{R}^2$ be a vector field, and let C be a planar curve with unit tangent vector $\hat{T}$ (recall §2.4). The *flux* of $\boldsymbol{F}$ across C is defined as the line integral

$$\int_C \boldsymbol{F} \cdot \hat{\mathrm{n}}\,\mathrm{d}s, \tag{33}$$

where $\hat{\mathrm{n}}$ denotes the *unit normal vector* to C obtained by rotating $\hat{T}$ *clockwise* by 90 degrees.

Similarly to work, which according to (19) is obtained by summing the *tangential* component of the vector field, the flux is obtained by summing the *normal* component of the field. Though conceptually analogous, these two notions turn out to have very different physical interpretations. For instance, if $\boldsymbol{F}$ is a velocity field of some fluid, then the flux measures how much fluid passes through C per unit time. Note that the unit normal vector $\hat{\mathrm{n}}$ points to the right as you walk along the curve C, and that whatever flows across C *left-to-right* (resp. *right-to-left*) is counted *positively* (resp. *negatively*).

Example 31. *Let $\boldsymbol{F} = x\hat{\imath} + y\hat{\jmath}$, and let C denote the circle of radius $a > 0$ centred at the origin, oriented counterclockwise. We would like to compute the flux of $\boldsymbol{F}$ across C. The first observation is that the unit normal vector to C, denoted $\hat{\mathrm{n}}$, points* *outwards. Since* $\boldsymbol{F} \parallel \hat{\mathrm{n}}$*, we have that* $\boldsymbol{F} \cdot \hat{\mathrm{n}} = |\boldsymbol{F}| = a$ *on* C*. It follows that*

$$\int_C \boldsymbol{F} \cdot \hat{\mathrm{n}}\, \mathrm{d}s = \int_C a\, \mathrm{d}s = a\, \mathrm{length}(C) = 2\pi a^2. \tag{34}$$

As expected, this is a positive quantity since things are flowing out *of the circle.*

Example 32. *Let* C *still denote the circle of radius* $a > 0$ *centred at the origin, oriented counterclockwise, but let us now take* $\boldsymbol{F} = -y\hat{\imath} + x\hat{\jmath}$*. In this case, we have already seen that* $\boldsymbol{F}$ *is tangent to* C*, and so* $\boldsymbol{F} \cdot \hat{\mathrm{n}} = 0$ *on* C*. It follows that the flux is zero,*

$$\int_C \boldsymbol{F} \cdot \hat{\mathrm{n}}\, \mathrm{d}s = 0.$$

To express the flux in coordinates, recall that $\mathrm{d}\boldsymbol{r} = \hat{T}\, \mathrm{d}s = (\mathrm{d}x, \mathrm{d}y)$. Since $\hat{\mathrm{n}}$ is obtained from $\hat{T}$ by a clockwise rotation of 90 degrees,

$$\hat{\mathrm{n}}\, \mathrm{d}s = (\mathrm{d}y, -\mathrm{d}x).$$

Given a vector field $\boldsymbol{F} = (P, Q)$, we then have that

$$\int_C \boldsymbol{F} \cdot \hat{\mathrm{n}}\, \mathrm{d}s = \int_C (P, Q) \cdot (\mathrm{d}y, -\mathrm{d}x) = \int_C -Q\, \mathrm{d}x + P\, \mathrm{d}y.$$

There is a version of Green's Theorem for flux. In order to state it, we first need to introduce the notion of *divergence* of a vector field.

Definition 33. *Let* $\boldsymbol{F} = P\hat{\imath} + Q\hat{\jmath}$ *be a vector field in the plane. Its* **divergence** *is given by*

$$\mathrm{div}(\boldsymbol{F}) := P_x + Q_y.$$

Informally, the divergence measures the "source rate", i.e. the amount of fluid added to or subtracted from the system per unit time and unit area. We can now state Green's Theorem for flux, commonly known as Green's Theorem in Normal Form.

Theorem 34 (Green's Theorem in Normal Form)**.** *Let* C *be a closed curve, oriented counterclockwise, and enclosing a region* $R \subset \mathbb{R}^2$*. If* $\boldsymbol{F}$ *is a vector field defined and differentiable everywhere in* R*, then*

$$\oint_C \boldsymbol{F} \cdot \hat{\mathrm{n}}\, \mathrm{d}s = \iint_R \mathrm{div}(\boldsymbol{F})\, \mathrm{d}A. \tag{35}$$

Recall that the left-hand side of (35) measures the flux out of the region R through C. In coordinates $\boldsymbol{F} = (P, Q)$, identity (35) reads as follows:

$$\oint_C -Q\, \mathrm{d}x + P\, \mathrm{d}y = \iint_R (P_x + Q_y)\, \mathrm{d}A. \tag{36}$$

Considering a new vector field $\boldsymbol{G} = (M, N)$, where $M = -Q$ and $N = P$, we can equivalently rewrite (36) as

$$\oint_C M\,\mathrm{d}x + N\,\mathrm{d}y = \iint_R (N_x - M_y)\,\mathrm{d}A,$$

which is just the usual Green's Theorem for the vector field $\boldsymbol{G}$. So the two versions are equivalent, and we do not need to prove Theorem 34 again.

Example 35. *Let $\boldsymbol{F} = x\hat{\imath} + y\hat{\jmath}$, let R denote the disc of radius $a > 0$ centred at the origin, and let C denote its boundary (oriented counterclockwise). The divergence of the vector field $\boldsymbol{F}$ is given by*

$$\operatorname{div}(\boldsymbol{F}) = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) = 1 + 1 = 2.$$

By Green's Theorem in Normal Form,

$$\oint_C \boldsymbol{F} \cdot \hat{\mathrm{n}}\,\mathrm{d}s = \iint_R 2\,\mathrm{d}A = 2\,\mathit{area}(R) = 2\pi a^2,$$

in accordance with Example 31.

We remark that, in the particular case of Example 35 where the divergence of $\boldsymbol{F}$ is constant, Green's Theorem in Normal Form allows us to go one step further and conclude that the flux of $\boldsymbol{F}$ across a circle of fixed radius is *independent* of the location of the centre of the circle.

4.5. **Simply-connected regions.** Green's Theorems 28 and 34 require that the vector field $\boldsymbol{F}$ and some of its derivatives are defined *everywhere* in the region R. Indeed, if the vector field is undefined at one single point inside R, we may run into problems. The next example illustrates this.

Example 36. *Consider the vector field*

$$\boldsymbol{F} = \frac{-y\hat{\imath} + x\hat{\jmath}}{x^2 + y^2},$$

which is not defined at the origin, but satisfies $\operatorname{curl}(\boldsymbol{F}) = 0$ everywhere else. Indeed, setting $M(x, y) = -\frac{y}{x^2+y^2}$ and $N(x, y) = \frac{x}{x^2+y^2}$, we have that

$$N_x = \frac{y^2 - x^2}{(x^2 + y^2)^2} = M_y.$$

If C is a closed curve whose interior R does not contain the origin, then Green's Theorem applies and yields

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \iint_R \operatorname{curl}(\boldsymbol{F})\,\mathrm{d}A = \iint_R 0\,\mathrm{d}A = 0.$$

On the other hand, if C' denotes the unit circle centred at the origin transversed counterclockwise, then a straightforward computation reveals that

$$(37)\quad \oint_{C'} \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{r} = \int_0^{2\pi} -\frac{\sin\theta}{\cos^2\theta+\sin^2\theta}(-\sin\theta\,\mathrm{d}\theta) + \frac{\cos\theta}{\cos^2\theta+\sin^2\theta}(\cos\theta\,\mathrm{d}\theta)$$
$$= \int_0^{2\pi} 1\,\mathrm{d}\theta = 2\pi \neq 0,$$

and so one cannot apply Green's Theorem directly. If C'' is any other closed curve whose interior contains the origin (transversed counterclockwise), then Green's Theorem and (37) *together imply that*

$$\oint_{C''} \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{r} = \oint_{C'} \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{r} = 2\pi.$$

In this way, the value of the line integral is seen to be independent of the curve C'', provided the latter is chosen among the class of curves that contain the origin in its interior.

Definition 37. *A region $R \subset \mathbb{R}^2$ is* **connected** *if it consists of a single piece. A connected region $R \subset \mathbb{R}^2$ is* **simply-connected** *if the interior of any closed curve in R is also contained in R.*

Intuitively, a simply-connected region does not have any "holes" in it.

Example 38. *The domain of definition of the vector field*

$$\boldsymbol{F} = \frac{-y\hat{\imath} + x\hat{\jmath}}{x^2+y^2}$$

considered in the previous example coincides with the whole plane minus the origin, i.e. $\mathbb{R}^2 \setminus \{(0,0)\}$. This is not a simply-connected region of the plane.

It turns out that, if the domain of definition and differentiability of a given vector field $\boldsymbol{F}$ is simply-connected, then we can apply Green's Theorems 28 and 34. Similarly, if the domain of definition and differentiability of a given vector field $\boldsymbol{F}$ is simply-connected, and if $\operatorname{curl}(\boldsymbol{F}) \equiv 0$, then $\boldsymbol{F}$ is conservative.

5. Interlude: Triple Integrals

Triple integrals in *rectangular, cylindrical* and *spherical coordinates* have already been covered in 2MVAa. We shall revise them quickly, taking a somewhat different approach in some aspects of the theory that will be convenient later on.

5.1. **Rectangular and cylindrical coordinates.** Given a region $D \subset \mathbb{R}^3$ and a function $f : D \to \mathbb{R}$, we are interested in integrals of the form

$$\iiint_D f \, \mathrm{d}V,$$

where $\mathrm{d}V = \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$ denotes the volume element in $\mathbb{R}^3$.

Example 39. *Let us compute the volume of the region D enclosed by the paraboloids*

$$z = x^2 + y^2 \textit{ and } z = 4 - x^2 - y^2.$$

The volume is given by[11]

$$\mathrm{Vol}(D) = \iiint_D 1 \, \mathrm{d}V.$$

To set up the bounds of integration, the first observation is that for fixed (x, y), the variable z varies between $z_{\text{bottom}} = x^2 + y^2$ and $z_{\text{top}} = 4 - x^2 - y^2$. On the other hand, the *shadow* of the region D into the xy-plane corresponds to the set of points (x, y) for which $z_{\text{bottom}} \leq z_{\text{top}}$, i.e.

$$x^2 + y^2 \leq 4 - x^2 - y^2.$$

In turn, this can be rewritten as $x^2+y^2 \leq 2$, and thus corresponds to the disc centred at the origin of radius $\sqrt{2}$. In other words, for each $-\sqrt{2} \leq x \leq \sqrt{2}$, the range for admissible values of y is $[-\sqrt{2 - x^2}, \sqrt{2 - x^2}]$. We finally conclude that

$$\mathrm{Vol}(D) = \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{x^2+y^2}^{4-x^2-y^2} 1 \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x.$$

Evaluating the inner integral (in z), we obtain

$$\mathrm{Vol}(D) = \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} (4 - 2x^2 - 2y^2) \, \mathrm{d}y \, \mathrm{d}x.$$

The latter (double) integral turns out to be easier to evaluate in polar coordinates $(x, y) = (r \cos\theta, r \sin\theta)$:

$$\mathrm{Vol}(D) = \int_0^{2\pi} \int_0^{\sqrt{2}} (4 - 2r^2) r \, \mathrm{d}r \, \mathrm{d}\theta = 2\pi \left[2r^2 - \tfrac{r^4}{2} \right]_0^{\sqrt{2}} = 4\pi. \tag{38}$$

Note that the integral in (38) can be rewritten as

$$\mathrm{Vol}(D) = \int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^{4-r^2} r \, \mathrm{d}z \, \mathrm{d}r \, \mathrm{d}\theta,$$

[11] The quantity $\mathrm{Vol}(D)$ could alternatively be set up as a certain double integral, but here we are interested in setting up the triple integral.

in the so-called *cylindrical coordinates*:

$$(x, y, z) = (r\cos\theta, r\sin\theta, z). \tag{39}$$

Here $r \geq 0$, and θ is the angle measured from the x-axis counterclockwise when seen from above.

Cylindrical coordinates are useful in problems that involve symmetry about an *axis*, and we may choose the z-axis to coincide with this axis of symmetry. In cylindrical coordinates, the equation $r = a$ defines a cylinder centred around the z-axis with base diameter a (hence the name "cylindrical" coordinates). On the other hand, the equation $\theta = b$ defines a vertical half-plane that contains the z-axis. The volume element is given by

$$\mathrm{d}V = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z,$$

see Figure 5 below.

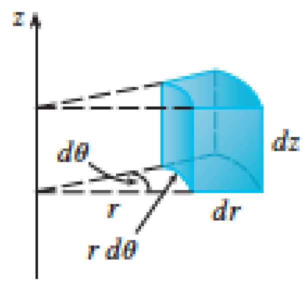

FIGURE 5. Volume element in cylindrical coordinates: $\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z$.

5.1.1. *Applications.* Triple integrals can be used for a number of purposes, including the computation of:

- *Volume*: As already mentioned, the volume of a region $D \subset \mathbb{R}^3$ is given by
$$\mathrm{Vol}(D) = \iiint_D 1\,\mathrm{d}V.$$
- *Mass*: The total mass of a solid object $D \subset \mathbb{R}^3$ of mass element $\mathrm{d}m$ and density $\delta = \frac{\mathrm{d}m}{\mathrm{d}V}$ is given by
$$\mathrm{Mass}(D) = \iiint_D \delta\,\mathrm{d}V.$$
In general, $\delta = \delta(x, y, z)$. In the special case when the density is constant, $\delta \equiv 1$, then the mass reduces to the volume, $\mathrm{Mass}(D) = \mathrm{Vol}(D)$.

- *Average Value:* Given a function $f : \mathbb{R}^3 \to \mathbb{R}$, its average value on a region $D \subset \mathbb{R}^3$ is given by
$$\bar{f} = \frac{1}{\mathrm{Vol}(D)} \iiint_D f \,\mathrm{d}V.$$
- *Centre of Mass*: The x-coordinate of the centre of mass of a solid object $D \subset \mathbb{R}^3$ of density $\delta = \delta(x, y, z)$ is given by
$$\bar{x} = \frac{1}{\mathrm{Mass}(D)} \iiint_D x\delta \,\mathrm{d}V.$$
Similarly,
$$\bar{y} = \frac{1}{\mathrm{Mass}(D)} \iiint_D y\delta \,\mathrm{d}V \text{ and } \bar{z} = \frac{1}{\mathrm{Mass}(D)} \iiint_D z\delta \,\mathrm{d}V.$$
- *Moment of Inertia*: The moment of inertia with respect to the z-axis of a solid object $D \subset \mathbb{R}^3$ of density $\delta = \delta(x, y, z)$ is given by
$$I_z = \iiint_D r^2\delta \,\mathrm{d}V = \iiint_D (x^2 + y^2)\delta \,\mathrm{d}V. \tag{40}$$
Similarly, the moments of inertia with respect to the x- and y-axes are respectively given by
$$I_x = \iiint_D (y^2 + z^2)\delta \,\mathrm{d}V \text{ and } I_y = \iiint_D (x^2 + z^2)\delta \,\mathrm{d}V.$$

Example 40. *Let $a, b > 0$, and consider the solid cone between the surfaces given in cylindrical coordinates by $z = ar$ and $z = b$. Assume constant density $\delta \equiv 1$. The moment of inertia of the cone with respect to the z-axis, denoted I_z as in* (40)*, measures how hard it is to spin the cone around the z-axis. It is given in cylindrical coordinates by*
$$I_z = \int_0^b \int_0^{2\pi} \int_0^{z/a} r^2 r \,\mathrm{d}r \,\mathrm{d}\theta \,\mathrm{d}z = \frac{\pi b^5}{10a^4}.$$

Example 41. *Consider the intersection of the region given in rectangular coordinates by $z \geq 1 - y$ and the unit ball centred at the origin, given as usual by $x^2 + y^2 + z^2 \leq 1$. Let D denote this intersection, and let us set up the triple integral corresponding to* $\mathrm{Vol}(D)$*.*

The first observation is that, for fixed $(x, y) \in \mathbb{R}^2$ satisfying $x^2 + y^2 \leq 1$, the variable z is constrained via the inequalities
$$1 - y \leq z \leq \sqrt{1 - x^2 - y^2}.$$

Secondly, the plane $z = 1 - y$ lies below the hemisphere $z = \sqrt{1 - x^2 - y^2}$ only if $1 - y \leq \sqrt{1 - x^2 - y^2}$, or equivalently

$$-\sqrt{2y - 2y^2} \leq x \leq \sqrt{2y - 2y^2}.$$

It follows that

$$\mathrm{Vol}(D) = \int_0^1 \int_{-\sqrt{2y-2y^2}}^{\sqrt{2y-2y^2}} \int_{1-y}^{\sqrt{1-x^2-y^2}} \mathrm{d}z \, \mathrm{d}x \, \mathrm{d}y.$$

This volume turns out to be easier to compute in spherical coordinates, and we will come back to it in Example 44 below.

5.2. **Spherical coordinates.** These are the three-dimensional analogue of polar coordinates. A generic point $(x, y, z) \in \mathbb{R}^3$ is represented in terms of the following parameters:

- *Distance from the origin*, denoted by $\rho \in [0, \infty)$;
- *Spherical angle*, measured down from the z-axis, denoted by $\phi \in [0, \pi]$;
- *Polar angle*, measured counterclockwise from the x-axis after projecting down to the xy-plane, denoted by $\theta \in [0, 2\pi]$ as in the case of polar coordinates.

The angles ϕ, θ are roughly analogous to *latitude* and *longitude*, respectively, even if the usual geographical and mathematical conventions do not always coincide. The key to switching from cylindrical coordinates (r, θ, z) to spherical coordinates (ρ, ϕ, θ) is to realise that that the latter correspond to polar coordinates in the rz-plane. Indeed,

$$z = \rho \cos \phi \text{ and } r = \rho \sin \phi, \tag{41}$$

whence we conclude from (39) that

$$\begin{aligned} x &= r \cos \theta = \rho \sin \phi \cos \theta \\ y &= r \sin \theta = \rho \sin \phi \sin \theta \\ z &= \rho \cos \phi. \end{aligned}$$

To switch backwards (from spherical to cylindrical coordinates), additionally note that

$$\rho = \sqrt{r^2 + z^2} = \sqrt{x^2 + y^2 + z^2}.$$

The spherical coordinate system is especially useful is problems where there is symmetry about a *point* (usually taken to be the origin).

Example 42. *The equation $\rho = a$ defines a sphere of radius a centred at the origin.*

Example 43. *The equation $\phi = \frac{\pi}{4}$ defines a cone of aperture $\frac{\pi}{2}$ centred around the z-axis with the vertex at the origin. Since $\cos(\frac{\pi}{4}) = \sin(\frac{\pi}{4}) = \frac{1}{\sqrt{2}}$, we see from (41) that in cylindrical coordinates this corresponds to the equation $z = r$.*

To compute triple integrals in spherical coordinates, we need to express the volume element $\mathrm{d}V$ in terms of $\mathrm{d}\rho, \mathrm{d}\phi$ and $\mathrm{d}\theta$. This amounts to a fun computation that involves realising that the area ΔS of a "spherical rectangle" corresponding to small angles $\Delta\phi$, $\Delta\theta$ is given by

$$\Delta S \simeq (\rho \sin\phi\, \Delta\theta)(\rho\, \Delta\phi) = \rho^2 \sin\phi\, \Delta\phi\, \Delta\theta,$$

where ρ denotes the radius of the corresponding sphere. Taking the limit as $\Delta\phi, \Delta\theta$ tend to zero, we obtain that the *surface area element* of a sphere of radius ρ is given by

$$\mathrm{d}S = \rho^2 \sin\phi\, \mathrm{d}\phi\, \mathrm{d}\theta. \tag{42}$$

Adding some depth to the picture, see Figure 6 below, we further have that

$$\Delta V \simeq (\Delta\rho)(\Delta S) = \rho^2 \sin\phi\, \Delta\rho\, \Delta\phi\, \Delta\theta,$$

and so

$$\mathrm{d}V = \rho^2 \sin\phi\, \mathrm{d}\rho\, \mathrm{d}\phi\, \mathrm{d}\theta.$$

This is an important formula which will be used many times and should be remembered by heart.

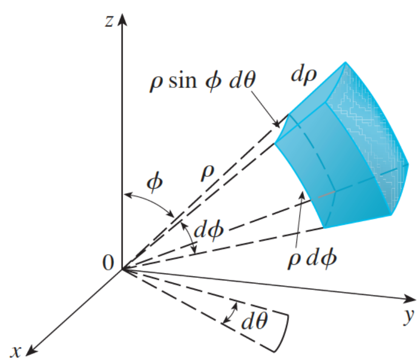

FIGURE 6. Volume element in spherical coordinates: $\mathrm{d}V = \rho^2 \sin\phi\, \mathrm{d}\rho\, \mathrm{d}\phi\, \mathrm{d}\theta$.

Example 44. *Consider the portion D of the unit ball centred at the origin which lies above the horizontal plane $z = \frac{1}{\sqrt{2}}$. Let us compute its volume,* $\mathrm{Vol}(D)$.

In spherical coordinates, the plane $z = \frac{1}{\sqrt{2}}$ is given by $\rho \cos\phi = \frac{1}{\sqrt{2}}$, or equivalently

$$\rho = \frac{1}{\sqrt{2}\cos\phi}.$$

By Pythagoras Theorem, the maximal value of ϕ is $\frac{\pi}{4}$, and so the desired volume is

$$\mathrm{Vol}(D) = \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_{\frac{1}{\sqrt{2}\cos\phi}}^{1} \rho^2 \sin\phi \, \mathrm{d}\rho \, \mathrm{d}\phi \, \mathrm{d}\theta = \tfrac{2\pi}{3}\left(1 - \tfrac{5}{4\sqrt{2}}\right).$$

As an exercise, convince yourself that the volume which we just computed coincides with the volume of the region considered in Example 41.

5.2.1. *Application: Gravitational attraction.* The gravitational force $\mathrm{d}\boldsymbol{F}$ exerted by a point mass $\mathrm{d}M$ at $(x, y, z) \in \mathbb{R}^3$ on a mass m at the origin has magnitude[12]

$$|\mathrm{d}\boldsymbol{F}| = \frac{Gm \, \mathrm{d}M}{\rho^2},$$

and (unit) direction given by

$$\mathrm{dir}(\mathrm{d}\boldsymbol{F}) = \frac{(x, y, z)}{\rho}.$$

Therefore the force $\mathrm{d}\boldsymbol{F}$ exerted by $\mathrm{d}M$ is given by

$$\mathrm{d}\boldsymbol{F} = \frac{Gm \, \mathrm{d}M}{\rho^3}(x, y, z), \tag{43}$$

in accordance with Newton's Law of Gravitation (17). Recalling that $\mathrm{d}M = \delta \, \mathrm{d}V$, we can integrate both side of (43) to obtain the total force exerted on m,

$$\boldsymbol{F} = Gm \iiint \frac{(x, y, z)}{\rho^3} \delta \, \mathrm{d}V.$$

Let us now assume that the solid can be positioned in such a way that the z-axis is an axis of symmetry. Then $\boldsymbol{F} = (0, 0, F)$, where

$$F = Gm \iiint \frac{z}{\rho^3} \delta \, \mathrm{d}V.$$

In spherical coordinates, this further simplifies to

$$F = Gm \iiint \frac{\rho\cos\phi}{\rho^3} \delta \rho^2 \sin\phi \, \mathrm{d}\rho \, \mathrm{d}\phi \, \mathrm{d}\theta = Gm \iiint \delta \cos\phi \sin\phi \, \mathrm{d}\rho \, \mathrm{d}\phi \, \mathrm{d}\theta.$$

From the latter expression, one could deduce *Newton's Shell Theorem*: A spherically symmetric body with uniform density $\delta \equiv 1$ affects external objects gravitationally as though all of its mass were concentrated at a point at its centre.

[12] Here $G \simeq 6.67408 \times 10^{-11} \mathrm{Nm}^2/\mathrm{kg}^2$ denotes the gravitational constant and $\rho = \sqrt{x^2 + y^2 + z^2}$.

6. Vector fields in space

A *vector field* in (three-dimensional) space is a function $\boldsymbol{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{k}$, where P, Q, R are functions of x, y, z. In other words, $\boldsymbol{F} : \mathbb{R}^3 \to \mathbb{R}^3$, $(x, y, z) \mapsto \boldsymbol{F}(x, y, z)$, where for each point $(x, y, z) \in \mathbb{R}^3$ in space, $\boldsymbol{F}(x, y, z)$ is a vector that depends on x, y, z. Examples include the velocity field of a fluid moving in three dimensions, or a force field like a gravitational, electric or magnetic field.

6.1. **Flux.** In the plane, we defined the flux of a two-dimensional vector field through a planar curve in (33). In three-dimensional space, we define the flux of a vector field $\boldsymbol{F} : \mathbb{R}^3 \to \mathbb{R}^3$ through a two-dimensional *surface* $S \subset \mathbb{R}^3$ as the surface integral

$$\iint_S \boldsymbol{F} \cdot \hat{n} \, \mathrm{d}S. \tag{44}$$

There are several things to clarify. Firstly, $\hat{n}$ denotes the *unit normal vector* to the surface S. There are two possible choices for $\hat{n}$, and fixing one of them amounts to fixing an *orientation* for the surface. Secondly, $\mathrm{d}S$ corresponds to the surface area element. In §6.2 below we will see how to compute surface area elements (and surface integrals) in general. The following piece of notation will also be useful:

$$\mathrm{dS} = \hat{n} \, \mathrm{d}S.$$

A preliminary observation which follows immediately from the definition of flux is the following: If the vector field $\boldsymbol{F}$ happens to be *tangent* to the surface S at every point of S, then the flux of $\boldsymbol{F}$ through S is equal to zero. In fact, in this case we have that $\boldsymbol{F} \cdot \hat{n} = 0$ on S, and so the integral in (44) vanishes.

If $\boldsymbol{F}$ is a velocity field, then the flux of $\boldsymbol{F}$ through a surface S measures the amount of matter (fluid or gas) that travels across S per unit time. This physical interpretation is similar to the one for the two-dimensional case seen in §4.4.

Example 45. *Let us compute the flux of the vector field $\boldsymbol{F} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ through the sphere of radius $a > 0$ centred at the origin, which we denote by S.*

The first observation is that the unit normal vector at a point $(x, y, z) \in S$ is given by

$$\hat{n} = \frac{1}{a}(x, y, z). \tag{45}$$

Since $\boldsymbol{F}$ is parallel to $\hat{n}$ and $|\hat{n}| = 1$, it follows that $\boldsymbol{F} \cdot \hat{n} = |\boldsymbol{F}| = \sqrt{x^2 + y^2 + z^2} = a$ if $(x, y, z) \in S$, and therefore the flux of $\boldsymbol{F}$ through S is given by

$$\iint_S \boldsymbol{F} \, \mathrm{dS} = \iint_S \boldsymbol{F} \cdot \hat{n} \, \mathrm{d}S = \iint_S a \, \mathrm{d}S = a \, \mathrm{area}(S) = a(4\pi a^2) = 4\pi a^3.$$

Example 46. *Let us now compute the flux of the vector field* $\boldsymbol{G} = z\hat{\mathrm{k}}$ *through the same sphere of radius* a *centred at the origin, still denoted by* S*.*

As noted above in (45), $\hat{\mathrm{n}} = (x, y, z)/a$, and so

$$\boldsymbol{G} \cdot \hat{\mathrm{n}} = \frac{z^2}{a}.$$

The flux of $\boldsymbol{G}$ through S is then given by

$$\iint_S \boldsymbol{G}\, \mathrm{dS} = \iint_S \boldsymbol{G} \cdot \hat{\mathrm{n}}\, \mathrm{d}S = \iint_S \frac{z^2}{a}\, \mathrm{d}S.$$

To compute the latter integral, we recall from (42) that

$$\mathrm{d}S = a^2 \sin\phi\, \mathrm{d}\phi\, \mathrm{d}\theta,$$

where ϕ, θ are the latitude and longitude angles on the sphere, respectively. From (41) we know that $z = a\cos\phi$, and so the flux of $\boldsymbol{G}$ through S is given by

$$\iint_S \frac{z^2}{a}\, \mathrm{d}S = \int_0^{2\pi} \int_0^{\pi} \frac{a^2 \cos^2\phi}{a} a^2 \sin\phi\, \mathrm{d}\phi\, \mathrm{d}\theta = 2\pi a^3 \left[-\tfrac{1}{3}\cos^3\phi\right]_0^{\pi} = \frac{4}{3}\pi a^3. \tag{46}$$

The fact that this particular flux coincides with the volume of a ball of radius a is *not* a coincidence, as will be explained in §7.2 below.

6.2. **Surface area element and surface integrals.** We start by computing the surface area element $\mathrm{d}S$ in various concrete examples.

Example 47. *Consider the horizontal plane given by the equation* $z = a$*. Then* $\hat{\mathrm{n}} = \pm\hat{\mathrm{k}}$*, and* $\mathrm{d}S = \mathrm{d}x\, \mathrm{d}y$*. In a similar way, for the vertical plane* $x = a$ *(parallel to the* yz*-plane), we have that* $\hat{\mathrm{n}} = \pm\hat{\imath}$ *and* $\mathrm{d}S = \mathrm{d}y\, \mathrm{d}z$*, and for the vertical plane* $y = a$ *(parallel to the* xz*-plane), we have that* $\hat{\mathrm{n}} = \pm\hat{\jmath}$ *and* $\mathrm{d}S = \mathrm{d}x\, \mathrm{d}z$*.*

Example 48. *Consider the sphere of radius* $a > 0$ *centred at the origin. We have already seen that* $\hat{\mathrm{n}} = \pm(x, y, z)/a$ *and that*

$$\mathrm{d}S = a^2 \sin\phi\, \mathrm{d}\phi\, \mathrm{d}\theta. \tag{47}$$

Example 49. *Consider the cylinder of radius* $a > 0$ *around the* z*-axis, given by the equation* $x^2 + y^2 = a^2$*. In this case,* $\hat{\mathrm{n}} = \pm(x, y, 0)/a$ *and* $\mathrm{d}S = a\, \mathrm{d}\theta\, \mathrm{d}z$*.*

Example 50. *Let* S *be the surface given by the graph of a function* $f : \mathbb{R}^2 \to \mathbb{R}$*, i.e.* $z = f(x, y)$*. In this case, we have that*

$$\hat{\mathrm{n}}\, \mathrm{d}S = \pm(-f_x, -f_y, 1)\, \mathrm{d}x\, \mathrm{d}y. \tag{48}$$

This is an important formula that should be remembered by heart. In particular, we have the following useful consequence,

$$\mathrm{d}S = \sqrt{f_x^2 + f_y^2 + 1}\,\mathrm{d}x\,\mathrm{d}y,$$

which allows for the computation of various areas and surface integrals.

Let us derive formula (48). With that purpose in mind, consider a small square on the xy-plane with sides parallel to the coordinate axes, having one vertex at a given point $(x, y) \in \mathbb{R}^2$, and side lengths Δx and Δy. The *lift* of the point (x, y) to the surface S is simply the point $(x, y, f(x, y))$. By lifting the whole square to S, we obtain a region of area ΔS that is well-approximated by a parallelogram of sides $\boldsymbol{u}$ and $\boldsymbol{v}$, given by

$$\begin{aligned}\boldsymbol{u} &\simeq (x+\Delta x, y, f(x+\Delta x, y)) - (x, y, f(x, y)) = (\Delta x, 0, f(x+\Delta x, y) - f(x, y)),\\ \boldsymbol{v} &\simeq (x, y+\Delta y, f(x, y+\Delta y)) - (x, y, f(x, y)) = (0, \Delta y, f(x, y+\Delta y) - f(x, y)).\end{aligned}$$

If $\Delta x, \Delta y$ are small enough, then Taylor's formula implies

$$\begin{aligned}f(x+\Delta x, y) - f(x, y) &\simeq f_x(x, y)\Delta x,\\ f(x, y+\Delta y) - f(x, y) &\simeq f_y(x, y)\Delta y.\end{aligned}$$

It follows that $\boldsymbol{u} \simeq (1, 0, f_x)\,\Delta x$ and $\boldsymbol{v} \simeq (0, 1, f_y)\,\Delta y$, and therefore

$$\hat{\mathrm{n}}\,\Delta S \simeq \boldsymbol{u} \times \boldsymbol{v} \simeq \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{\mathrm{k}} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix} \Delta x\,\Delta y = (-f_x, -f_y, 1)\,\Delta x\,\Delta y,$$

where the unit normal vector $\hat{\mathrm{n}}$ is chosen to be pointing upwards. Letting $\Delta x, \Delta y$ tend to zero, we obtain (48).

Example 51. *Let us compute the flux of the vector field $\boldsymbol{F} = z\hat{\mathrm{k}}$ through the portion S of the paraboloid $z = x^2 + y^2$ which lies above the unit disc centred at the origin.*

We start by noting that this flux should be *positive* if we choose the unit normal vector $\hat{\mathrm{n}}$ to be pointing upwards (i.e. with a positive z-component). Appealing to formula (48) with $f(x, y) = x^2 + y^2$, we then have that

$$\hat{\mathrm{n}}\,\mathrm{d}S = (-2x, -2y, 1)\,\mathrm{d}x\,\mathrm{d}y.$$

Consequently,

$$\iint_S \boldsymbol{F} \cdot \hat{\mathrm{n}}\,\mathrm{d}S = \iint_S (0, 0, z) \cdot (-2x, -2y, 1)\,\mathrm{d}x\,\mathrm{d}y = \iint_S z\,\mathrm{d}x\,\mathrm{d}y.$$

On S we have that $z = x^2 + y^2$, and so

$$\iint_S z \, \mathrm{d}x \, \mathrm{d}y = \iint_S (x^2 + y^2) \, \mathrm{d}x \, \mathrm{d}y = \int_0^{2\pi} \int_0^1 r^2 r \, \mathrm{d}r \, \mathrm{d}\theta = \frac{\pi}{2}.$$

More generally, we can consider the case in which the surface S is given by parametric equations

$$x = x(u, v), \ y = y(u, v), \ z = z(u, v).$$

To determine $\hat{\mathrm{n}} \, \mathrm{d}S$ in terms of the parameters u, v (and the corresponding differentials $\mathrm{d}u$, $\mathrm{d}v$), note that the sides of the lift of a small square to S can be approximated in terms of the position vector $\boldsymbol{r} = \boldsymbol{r}(u, v) = (x(u, v), y(u, v), z(u, v))$ as follows:

$$\boldsymbol{r}_u \, \Delta u = (x_u, y_u, z_u) \, \Delta u, \quad \text{and} \quad \boldsymbol{r}_v \, \Delta v = (x_v, y_v, z_v) \, \Delta v.$$

Reasoning as before,

$$\pm \hat{\mathrm{n}} \, \Delta S \simeq (\boldsymbol{r}_u \, \Delta u \times \boldsymbol{r}_v \, \Delta v) = (\boldsymbol{r}_u \times \boldsymbol{r}_v) \, \Delta u \, \Delta v,$$

and letting $\Delta u, \Delta v$ tend to zero, we finally obtain

$$\hat{\mathrm{n}} \, \mathrm{d}S = \pm (\boldsymbol{r}_u \times \boldsymbol{r}_v) \, \mathrm{d}u \, \mathrm{d}v. \tag{49}$$

In particular,

$$\mathrm{d}S = |\boldsymbol{r}_u \times \boldsymbol{r}_v| \, \mathrm{d}u \, \mathrm{d}v,$$

a useful formula when it comes to computing various areas and surface integrals.

Formula (49) recovers all the previous formulas which we have obtained for $\hat{\mathrm{n}} \, \mathrm{d}S$. As an exercise, convince yourself that formula (47) with

$$(x(\phi, \theta), y(\phi, \theta), z(\phi, \theta)) = (a \sin\phi \cos\theta, a \sin\phi \sin\theta, a \cos\phi),$$

and formula (48) with

$$(x(u, v), y(u, v), z(u, v)) = (u, v, f(u, v)),$$

both follow from (49).

We discuss one last formula for $\hat{\mathrm{n}} \, \mathrm{d}S$. There are situations in which one can easily determine a normal vector $\boldsymbol{N}$ to a given surface S. Specific examples include:

(a) A normal vector to the plane $ax + by + cz = d$ is given by $\boldsymbol{N} = (a, b, c)$.
(b) A normal vector to the surface S defined by an equation $g(x, y, z) = 0$ is given by $\boldsymbol{N} = \nabla g$. This follows from the fact (proved in 2MVAa) that the gradient ∇g points in a direction perpendicular to the level surfaces of the function g.

If we happen to know a normal vector $\boldsymbol{N}$ (*not* necessarily of unit length) to a given surface S, then $\boldsymbol{n}\,\mathrm{d}S$ can be computed as follows.

Consider a small piece of surface of area ΔS with normal vector $\boldsymbol{N}$ which makes an angle α with the xy-plane. If ΔA denotes the area of its shadow on the xy-plane, then $\Delta A \simeq \Delta S \cos\alpha$. Note that $\boldsymbol{N}$ and $\hat{\mathrm{k}}$ are normal to the small piece of surface and to its shadow, respectively. Since the angle between two planes equals the angle between the corresponding normal vectors, we have that

$$\cos\alpha = \frac{\boldsymbol{N}\cdot\hat{\mathrm{k}}}{|\boldsymbol{N}|}.$$

It follows that

$$\Delta S \simeq \frac{1}{\cos\alpha}\Delta A = \frac{|\boldsymbol{N}|}{\boldsymbol{N}\cdot\hat{\mathrm{k}}}\Delta A.$$

Let $\hat{\mathrm{n}}$ denote the unit normal vector to S as usual. Since $\boldsymbol{N}$ and $\hat{\mathrm{n}}$ are collinear, we have that $|\boldsymbol{N}|\,\hat{\mathrm{n}} = \pm\boldsymbol{N}$, and therefore

$$\hat{\mathrm{n}}\,\Delta S \simeq \frac{|\boldsymbol{N}|\,\hat{\mathrm{n}}}{\boldsymbol{N}\cdot\hat{\mathrm{k}}}\Delta A = \pm\frac{\boldsymbol{N}}{\boldsymbol{N}\cdot\hat{\mathrm{k}}}\Delta A.$$

Letting ΔA tend to zero, we finally have that

$$\hat{\mathrm{n}}\,\mathrm{d}S = \pm\frac{\boldsymbol{N}}{\boldsymbol{N}\cdot\hat{\mathrm{k}}}\,\mathrm{d}A.$$

Example 52. *Let S be the surface given by the graph of a function, $z = f(x,y)$. Define the auxiliary function $g(x,y,z) := z - f(x,y)$. Then S is defined by the equation $g(x,y,z) = 0$, and so a normal vector to S is given by $\boldsymbol{N} = \nabla g$. But $\nabla g = (-f_x, -f_y, 1)$, and so*

$$\hat{\mathrm{n}}\,\mathrm{d}S = \frac{\boldsymbol{N}}{\boldsymbol{N}\cdot\hat{\mathrm{k}}}\,\mathrm{d}A = \frac{(-f_x,-f_y,1)}{(-f_x,-f_y,1)\cdot(0,0,1)}\,\mathrm{d}x\,\mathrm{d}y = (-f_x,-f_y,1)\,\mathrm{d}x\,\mathrm{d}y$$

if $\hat{\mathrm{n}}$ is chosen to point upwards. In this way, we recover formula (48) *once again.*

7. Divergence Theorem

We start by defining the *divergence* of a vector field in three-dimensional space. The modifications with respect to the two-dimensional version (Definition 33) are the natural ones.

Definition 53. *Let $\boldsymbol{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{\mathrm{k}}$ be a vector field in space. Its* **divergence** *is given by*

$$\mathrm{div}(\boldsymbol{F}) := P_x + Q_y + R_z.$$

If we let $\boldsymbol{F}$ be the velocity field of some fluid, then the divergence of $\boldsymbol{F}$ measures the *source rate*, i.e. the amount of flux generated by unit time and unit volume.

7.1. **Statement.** Gauss's Divergence Theorem is a result that relates the flux of a vector field through a surface to the behaviour of the field inside the surface.

Theorem 54 (Gauss' Divergence Theorem)**.** *Let S be a closed surface, oriented outwards, and enclosing a solid region $D \subset \mathbb{R}^3$. If $\boldsymbol{F}$ be a vector field defined and differentiable in D, then*

$$\iint_S \boldsymbol{F} \cdot \mathrm{dS} = \iiint_D \mathrm{div}(\boldsymbol{F})\, \mathrm{d}V. \tag{50}$$

Gauss' Divergence Theorem, sometimes called the Gauss–Green Theorem, is a three-dimensional analogue of Green's Theorem in Normal Form (Theorem 34). The integral notation on the left-hand side of (50) is meant to emphasise the fact that we are considering the flux through a *closed* surface S. This is a crucial assumption without which the Divergence Theorem cannot be applied!

Recall that $\mathrm{dS} = \hat{\mathrm{n}}\, \mathrm{d}S$, where $\hat{\mathrm{n}}$ denotes the outwards pointing unit normal vector. In coordinates $\boldsymbol{F} = (P, Q, R)$, identity (50) then reads as follows:

$$\iint_S (P, Q, R) \cdot \hat{\mathrm{n}}\, \mathrm{d}S = \iiint_D (P_x + Q_y + R_z)\, \mathrm{d}V.$$

The physical interpretation of Gauss' Divergence Theorem is better described in the particular case when $\boldsymbol{F}$ is the velocity field of some incompressible fluid (say, water). Then the left-hand side of (50) measures the amount of fluid leaving the region D per unit time, whereas the right-hand side of (50) measures[13] the number of sources minus the number of sinks. Thus is a certain sense the Divergence Theorem quantifies the intuitive principle that *all matter leaving a certain a region must come from somewhere.*

Notation. We pause to introduce a useful piece of notation, by defining the symbolic vector

$$\nabla := \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right), \tag{51}$$

(the symbol ∇ is pronounced as"nabla" and the result of application is sometimes pronounced as the "del" operator). The gradient of a function $f : \mathbb{R}^3 \to \mathbb{R}$ is then given by

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right),$$

[13]Recall the physical interpretation of the divergence given after Definition 53.

and the divergence of a vector field $\boldsymbol{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{\mathrm{k}}$ is given by

$$\nabla \cdot \boldsymbol{F} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \cdot (P, Q, R) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = \mathrm{div}(\boldsymbol{F}).$$

This notation will be especially useful when we define the *curl* of a three-dimensional vector field in §8.2 below.

7.2. **Revisiting an old example.** We return to Example 46. Consider the vector field $\boldsymbol{G} = z\hat{\mathrm{k}}$. Let S denote the sphere of radius a centred at the origin with outwards pointing unit normal $\hat{\mathrm{n}}$, and enclosing the ball D. Start by noting that

$$\mathrm{div}(\boldsymbol{G}) = \mathrm{div}(z\hat{k}) = 0 + 0 + 1 = 1.$$

From the Divergence Theorem, it readily follows that

$$\iint_S z\hat{\mathrm{k}} \cdot \hat{\mathrm{n}}\, \mathrm{d}S = \iiint_D \mathrm{div}(z\hat{\mathrm{k}})\, \mathrm{d}V = \iiint_D 1\, \mathrm{d}V = \mathrm{Vol}(D) = \frac{4}{3}\pi a^3,$$

in accordance with (46).

7.3. **Proof.** In this section we prove Gauss' Divergence Theorem. As we shall see, the proof follows the general scheme of that of Green's Theorem (which you may wish to recall before reading the upcoming proof).

Proof of Gauss' Divergence Theorem. We focus on the special case of a vector field $\boldsymbol{F} = (0, 0, R)$, for which we want to prove that

$$\iint_S (0, 0, R) \cdot \hat{\mathrm{n}}\, \mathrm{d}S = \iiint_D R_z\, \mathrm{d}V. \tag{52}$$

The general case will then follow from summing three such identities, one for each component P, Q, R. If the region D happens to be *vertically simple*, i.e. of the form

$$D = \{(x, y, z) \in \mathbb{R}^3 : (x, y) \in U \text{ and } z_1(x, y) \leq z \leq z_2(x, y)\}$$

for some domain $U \subset \mathbb{R}^2$ and functions $z_1, z_2 : \mathbb{R}^2 \to \mathbb{R}$, then we may proceed as follows. By the Fundamental Theorem of Calculus, the right-hand side of (52) can be simplified to

$$\begin{aligned}\iiint_D R_z\, \mathrm{d}V &= \iint_U \left(\int_{z_1(x,y)}^{z_2(x,y)} R_z\, \mathrm{d}z\right) \mathrm{d}x\, \mathrm{d}y \\ &= \iint_U \left(R(x, y, z_2(x, y)) - R(x, y, z_1(x, y))\right) \mathrm{d}x\, \mathrm{d}y.\end{aligned}$$

On the other hand, the left-hand side of (52) equals the flux through a surface which contains three distinct components: *top*, *bottom* and *sides*. The top is given by the graph of the function $z = z_2(x, y)$, and as such it follows[14] from (48) that

$$\hat{\mathrm{n}}\,\mathrm{d}S = \left(-\frac{\partial z_2}{\partial x}, -\frac{\partial z_2}{\partial y}, 1\right)\,\mathrm{d}x\,\mathrm{d}y.$$

In particular, $(0, 0, R) \cdot \hat{\mathrm{n}}\,\mathrm{d}S = R\,\mathrm{d}x\,\mathrm{d}y$, and so

$$\iint_{\mathrm{top}} (0, 0, R) \cdot \hat{\mathrm{n}}\,\mathrm{d}S = \iint_{\mathrm{top}} R\,\mathrm{d}x\,\mathrm{d}y = \iint_U R(x, y, z_2(x, y))\,\mathrm{d}x\,\mathrm{d}y.$$

In a similar way, we check that on the bottom[15]

$$\hat{\mathrm{n}}\,\mathrm{d}S = \left(\frac{\partial z_1}{\partial x}, \frac{\partial z_1}{\partial y}, -1\right)\,\mathrm{d}x\,\mathrm{d}y,$$

and therefore

$$\iint_{\mathrm{bottom}} (0, 0, R) \cdot \hat{\mathrm{n}}\,\mathrm{d}S = \iint_{\mathrm{bottom}} -R\,\mathrm{d}x\,\mathrm{d}y = \iint_U -R(x, y, z_1(x, y))\,\mathrm{d}x\,\mathrm{d}y.$$

Finally, since the sides are vertical, the vector $(0, 0, R)$ is tangent to the sides, and therefore the flux through the sides vanishes (without the need for further computations). This verifies identity (52) in the particular case of a vertically simple region D.

If D is not vertically simple, then we may decompose it into finitely many vertically simple pieces. Applying the result for each of them, and then summing the various contributions, finishes the proof. □

7.4. **Application: diffusion equation.** The *diffusion equation* is a partial differential equation that governs the motion of e.g. smoke in immobile air, or dye in a given solution. The unknown is a function $u = u(x, y, z, t)$ of *four* variables that measures the concentration of matter (smoke, dye, ...) at a given point $(x, y, z) \in \mathbb{R}^3$ and at time $t \geq 0$. The diffusion equation reads as follows:

$$\frac{\partial u}{\partial t} = k\Delta u, \tag{53}$$

where Δu is the *Laplacian* of u, a second-order differential operator defined as

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}.$$

[14] Recall that S is oriented outwards by assumption. In particular, $\hat{\mathrm{n}}$ points *upwards* on top.

[15] Note that now $\hat{\mathrm{n}}$ points *downwards*.

The Laplacian of u equals the divergence of the gradient of u,

$$\begin{aligned}\operatorname{div}(\nabla u) &= \operatorname{div}\left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right) \\ &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial u}{\partial z}\right) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \Delta u,\end{aligned}$$

and can thus be written in terms of the notation introduced in (51) as $\Delta u = \nabla \cdot (\nabla u)$.

Let us try to understand the reason why equation (53) turns out to be a reasonable model for, say, the diffusion of smoke. Consider the vector field $\boldsymbol{F}$ corresponding to the velocity field of the smoke. Since smoke flows from regions of high concentration to regions of low concentration, we see that $\boldsymbol{F}$ is directed along the gradient[16] $-\nabla u$. In fact, these two quantities are known to be proportional:

$$\boldsymbol{F} = -k\nabla u, \text{ for some } k > 0.$$

Now, given a certain solid region $D \subset \mathbb{R}^3$ with boundary S, the flux of smoke out of D through S measures the amount of matter through S per unit time. Since concentration measures the amount of matter per unit volume, we have that

$$\iint_S \boldsymbol{F} \cdot \hat{\mathrm{n}} \, \mathrm{d}S = -\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_D u \, \mathrm{d}V\right). \tag{54}$$

The minus sign on the right-hand side of (54) is due to the choice of (outwards pointing) unit normal vector $\hat{\mathrm{n}}$. By the Divergence Theorem, we have that

$$\iint_S \boldsymbol{F} \cdot \hat{\mathrm{n}} \, \mathrm{d}S = \iiint_D \operatorname{div}(\boldsymbol{F}) \, \mathrm{d}V.$$

It follows that

$$\iiint_D \operatorname{div}(\boldsymbol{F}) \, \mathrm{d}V = -\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_D u \, \mathrm{d}V\right) = -\iiint_D \frac{\partial u}{\partial t} \, \mathrm{d}V,$$

and so

$$\frac{1}{\mathrm{Vol}(D)} \iiint_D \operatorname{div}(\boldsymbol{F}) \, \mathrm{d}V = \frac{1}{\mathrm{Vol}(D)} \iiint_D -\frac{\partial u}{\partial t} \, \mathrm{d}V.$$

In other words, the average values on D of the functions $\operatorname{div}(\boldsymbol{F})$ and $-\frac{\partial u}{\partial t}$ are the same, *for any region* $D \subset \mathbb{R}^3$. Therefore the two functions coincide,

$$\operatorname{div}(\boldsymbol{F}) = -\frac{\partial u}{\partial t}.$$

[16] The minus sign is there because the flow goes from high to low concentration regions (and not vice-versa).

To conclude the derivation of (53), simply recall that $\boldsymbol{F} = -k\nabla u$, and note that in this case

$$\operatorname{div}(\boldsymbol{F}) = \operatorname{div}(-k\nabla u) = -k\operatorname{div}(\nabla u) = -k\Delta u.$$

8. Stokes' Theorem

8.1. **Line integrals in space.** Line integrals in three-dimensional space are conceptually analogous to their two-dimensional variants studied in §3.1. Given a vector field $\boldsymbol{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{k}$ and a curve C in $\mathbb{R}^3$, the work done by $\boldsymbol{F}$ along C is given by the line integral

$$\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_C (P, Q, R) \cdot (\mathrm{d}x, \mathrm{d}y, \mathrm{d}z) = \int_C P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z.$$

To evaluate such an integral, we parametrise the curve C, and then express the variables x, y, z (together with $\mathrm{d}x, \mathrm{d}y, \mathrm{d}z$) in terms of the chosen parameter.

Example 55. *Let us compute the line integral of the vector field* $\boldsymbol{F} = (yz, xz, xy)$ *along the curve* C *given by*

$$C : (x, y, z) = (t^3, t^2, t), \;\; 0 \le t \le 1.$$

Since $\mathrm{d}x = 3t^2\,\mathrm{d}t$, $\mathrm{d}y = 2t\,\mathrm{d}t$ and $\mathrm{d}z = \mathrm{d}t$, we have that

$$\begin{aligned}\int_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} &= \int_C yz\,\mathrm{d}x + xz\,\mathrm{d}y + xy\,\mathrm{d}z \\ &= \int_C t^3(3t^2\,\mathrm{d}t) + t^4(2t\,\mathrm{d}t) + t^5\,\mathrm{d}t = \int_0^1 6t^5\,\mathrm{d}t = [t^6]_0^1 = 1.\end{aligned}$$

Example 56. *Let us consider the same vector field* $\boldsymbol{F}$ *as in Example 55, but now the curve* $C' = C_1 + C_2 + C_3$ *will consist of three line segments:* C_1 *joins the points* $(0, 0, 0)$ *and* $(1, 0, 0)$*,* C_2 *joins the points* $(1, 0, 0)$ *and* $(1, 1, 0)$*, and* C_3 *joins the points* $(1, 1, 0)$ *and* $(1, 1, 1)$*.*

Since C_1 and C_2 lie on the xy-plane, for both of them we have that $z = 0 = \mathrm{d}z$. Therefore the corresponding line integrals vanish. On C_3, we have that $x = 1 = y$ and therefore $\mathrm{d}x = 0 = \mathrm{d}y$. It follows that

$$\int_{C'} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \left(\int_{C_1} + \int_{C_2} + \int_{C_3}\right) \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = 0 + 0 + \int_{C_3} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_{C_3} xy\,\mathrm{d}z = \int_0^1 1\,\mathrm{d}z = 1.$$

This result could have been anticipated, by noting that the line integral in question is path-independent since $\boldsymbol{F}$ is a gradient field. In fact, defining $f(x, y, z) = xyz$, one readily checks that $\boldsymbol{F} = \nabla f$. It then follows by the Fundamental Theorem of Calculus for Line Integrals[17] that

$$\int_{C'} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_{C'} \nabla f \cdot \mathrm{d}\boldsymbol{r} = f(1,1,1) - f(0,0,0) = 1 - 0 = 1.$$

So far the two- and three-dimensional theories seem to be almost perfect analogues of each other. One place where things start to differ is when it comes to testing whether a certain vector field is a gradient field or not. If a vector field $\boldsymbol{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{\mathrm{k}}$ happens to coincide with the gradient of a function $\nabla f = (f_x, f_y, f_z)$, then we have that[18]

$$\begin{aligned} P_y &= f_{xy} = f_{yx} = Q_x \\ P_z &= f_{xz} = f_{zx} = R_x \\ Q_z &= f_{yz} = f_{zy} = R_y. \end{aligned}$$

It follows that a vector field $\boldsymbol{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{\mathrm{k}}$ defined on the whole space $\mathbb{R}^3$ is a gradient field *if and only if*

$$P_y = Q_x \text{ and } P_z = R_x \text{ and } Q_z = R_y.$$

Example 57. *For which $a, b \in \mathbb{R}$ is*

$$\boldsymbol{F} = axy\,\hat{\imath} + (x^2 + z^3)\,\hat{\jmath} + (byz^2 - 4z^3)\,\hat{\mathrm{k}}$$

a gradient field?

From $ax = P_y = Q_x = 2x$, we conclude that $a = 2$. From $3z^2 = Q_z = R_y = bz^2$, we conclude that $b = 3$. Additionally note that the remaining identity $0 = P_z = R_x = 0$ does not give any extra information.

For the values $(a, b) = (2, 3)$ which we just determined, let us try to determine the corresponding potential f. We may use any of the two different methods described in §3.3 for the two-dimensional situation:

Method 1. *Computing line integrals.* This works in exactly the same way as in the two-dimensional case (one just has to compute three line integrals instead of just two). We omit the details.

[17]Note that the proof of Theorem 19 applies equally well to the three-dimensional setting.

[18]Here we assume that the second-order partial derivatives of f are continuous, so that we can appeal to Clairaut's Theorem to ensure that the mixed partial derivatives coincide.

Method 2. *Computing anti-derivatives.* Let us find a function $f : \mathbb{R}^3 \to \mathbb{R}$, such that

$$f_x = 2xy \text{ and } f_y = x^2 + z^3 \text{ and } f_z = 3yz^2 - 4z^3. \tag{55}$$

We may integrate the first identity in (55) with respect to the variable x, and conclude that $f(x, y, z) = x^2y + g(y, z)$ for some function g that does *not* depend on x but may depend on y, z. Differentiating this with respect to y, we conclude that

$$f_y(x, y, z) = x^2 + g_y(y, z) = x^2 + z^3,$$

where the latter identity follows from the second identity in (55). In particular, $g_y(y, z) = z^3$, and so $g(y, z) = yz^3 + h(z)$, for some function h which is independent of the variables x, y but may depend on z. We conclude that

$$f(x, y, z) = x^2y + yz^3 + h(z),$$

which can be differentiated with respect to z to yield

$$f_z(x, y, z) = 3yz^2 + h'(z) = 3yz^2 - 4z^3.$$

Here the last identity follows from the third identity in (55). Consequently $h'(z) = -4z^3$, and so $h(z) = -z^4 + C$, for some (proper) constant C. All in all, we have that

$$f(x, y, z) = x^2y + yz^3 - z^4 + C.$$

8.2. **Curl.** We introduce the notion of *curl* of a vector field in space, which is a measure of how far a given vector field is from being conservative.

Definition 58. *Let* $\boldsymbol{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{k}$ *be a vector field in space. Its* **curl** *is the vector given by*

$$\operatorname{curl}(\boldsymbol{F}) := (R_y - Q_z)\hat{\imath} + (P_z - R_x)\hat{\jmath} + (Q_x - P_y)\hat{k}. \tag{56}$$

Note carefully that the curl of a three-dimensional vector field is itself a *vector*, as opposed to the curl of a two-dimensional vector field which is a scalar (recall Definition 24).

The notation $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$ introduced in (51) provides a useful way to quickly derive the components of the curl of a vector field $\boldsymbol{F} = (P, Q, R)$:

$$\nabla \times \boldsymbol{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = (R_y - Q_z)\hat{\imath} - (R_x - P_z)\hat{\jmath} + (Q_x - P_y)\hat{k} = \operatorname{curl}(\boldsymbol{F}).$$

From our previous discussion, it follows that the curl of a vector field $\boldsymbol{F}$ which is defined everywhere vanishes if and only if $\boldsymbol{F}$ is conservative.

The curl of a vector field has actual physical meaning. For instance, the curl of a velocity field measures the rotation component of the motion. As a concrete example, take the vector field $\boldsymbol{v}(x,y,z) = (-\omega y, \omega x, 0)$ which corresponds to rotation about the z-axis at angular velocity ω. Then

$$\operatorname{curl}(\boldsymbol{v}) = \nabla \times \boldsymbol{v} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{\mathrm{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\omega y & \omega x & 0 \end{vmatrix} = 2\omega\hat{\mathrm{k}}. \tag{57}$$

Example 59. *The* curl *of a constant vector field* $\boldsymbol{F} = (a,b,c)$ *is the zero vector, as one would expect.*

Example 60. *Let* $\boldsymbol{F} = (x,0,0)$*. Then* $\operatorname{curl}(\boldsymbol{F}) = \mathbf{0}$ *but* $\operatorname{div}(\boldsymbol{F}) = 1$*, in line with the fact that the vector field* $\boldsymbol{F}$ *leads to stretching effects but no rotation effects.*

Example 61. *Let* $\boldsymbol{F} = (-y,x,0)$*. Then* $\operatorname{curl}(\boldsymbol{F}) = 2\hat{\mathrm{k}}$*, a vector which points along the* z*-axis of rotation with a magnitude that equals twice the angular velocity, in accordance with* (57)*.*

8.3. **Statement and idea of proof.** This classical Stokes' Theorem relates the surface integral of the curl of a vector field over a surface in three-dimensional space to the line integral of the vector field over its boundary. Numerous generalisations of this foundational result (e.g. to higher dimensions) are to be found in more advanced literature.

In Theorem 62 below, both the curve C and the surface S need to be oriented, and the orientations have to be chosen in a *compatible* way. Informally, this means that if a particle moves along the curve C in the positive direction (i.e. counterclockwise when seen from above), having the surface S to its left, then the unit normal vector $\hat{\mathrm{n}}$ to S must be pointing upwards. Alternatively, we can use the right-hand rule discussed in §1.3 instead: Take your right hand, let your index finger point in the positive direction of the curve C, and place your middle finger tangent to the surface S pointing towards its interior. Then your thumb will point in the direction of the unit normal vector $\hat{\mathrm{n}}$.

Theorem 62 (Stokes' Theorem)**.** *Let* $C \subset \mathbb{R}^3$ *be a closed curve, and let* S *be any surface bounded by* C*. Assume the orientations of* C *and* S *to be compatible. If* $\boldsymbol{F}$ *be a vector field defined and differentiable in* S*, then*

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \iint_S (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}} \, \mathrm{d}S. \tag{58}$$

Given the striking similarity between Stokes' Theorem and Green's Theorem, it is perhaps not surprising that the latter turns out to be a special case of the former. To make this assertion precise, let S be a (bounded) portion of the xy-plane. Denote its boundary curve by C, and orient it counterclockwise. Consider a vector field $\boldsymbol{F} = (P, Q, R)$. Since C is contained in the xy-plane, on C we have that $z = \mathrm{d}z = 0$, and therefore

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \oint_C (P, Q, R) \cdot (\mathrm{d}x, \mathrm{d}y, 0) = \oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y. \tag{59}$$

Since S is also contained in the xy-plane, and C is oriented counterclockwise when seen from above, we take $\hat{\mathrm{n}} = \hat{\mathrm{k}}$. From (59) and Stokes' Theorem, it then follows that

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_S (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{k}}\,\mathrm{d}S.$$

But $(\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{k}}$ is just the z-component of the curl of $\boldsymbol{F}$, which according to Definition 58 equals $Q_x - P_y$. It then follows that

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_S (Q_x - P_y)\,\mathrm{d}x\,\mathrm{d}y,$$

which is just the familiar statement of Green's Theorem! In other words, Green's Theorem is the special case of Stokes' Theorem restricted to the xy-plane.

Even if the rigorous proof of Stokes' Theorem lies beyond the scope of the present course, we can still indicate a few reasons why one could expect such a statement to hold:

- We have just seen that Stokes' Theorem holds if the curve C and the surface S both lie on the xy-plane. A similar reasoning takes care of the cases when C and S both lie on the yz-plane, or both lie on the xz-plane.
- As a matter of fact, Stokes' Theorem holds if the curve C and the surface S both lie on *any* given plane (not necessarily the coordinate planes). A proof of this statement uses the fact that work, flux and curl all make sense independently of the specific coordinate system which we choose to work on.
- Let S be any given surface in three-dimensional space, and denote its boundary curve by C. It is *intuitively clear*[19] that S can be decomposed into tiny, almost flat pieces. The sum of the work done by a given vector field $\boldsymbol{F}$ around each little piece equals the total work done along C. Similarly, the sum of the flux of $\boldsymbol{F}$ through each little piece equals the total flux through S.
- Therefore Stokes' Theorem for flat pieces implies the general form of Stokes' Theorem, and this concludes our rough sketch of how a rigorous proof could be constructed.

[19]This is the kind of expression that should be avoided in a rigorous mathematical proof!

Example 63. *Let us compute the work done by the vector field* $\boldsymbol{F} = z\hat{\imath} + x\hat{\jmath} + y\hat{\mathrm{k}}$ *around the unit circle* C *in the* xy*-plane centred at the origin, oriented counterclockwise when seen from above.*

We can do it directly. As usual, the first step it to parametrise C, e.g. via

$$C : \theta \mapsto (x(\theta), y(\theta), z(\theta)) = (\cos\theta, \sin\theta, 0), \quad 0 \le \theta \le 2\pi.$$

Since $z = \mathrm{d}z = 0$, we then have that

$$\begin{aligned}\oint_C \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{r} &= \oint_C z\,\mathrm{d}x + x\,\mathrm{d}y + y\,\mathrm{d}z = \oint_C x\,\mathrm{d}y = \int_0^{2\pi} \cos\theta\,(\cos\theta\,\mathrm{d}\theta)\\ &= \int_0^{2\pi} \cos^2\theta\,\mathrm{d}\theta = \int_0^{2\pi} \frac{1+\cos(2\theta)}{2}\,\mathrm{d}\theta = \pi. \qquad (60)\end{aligned}$$

Alternatively, we can use Stokes' Theorem. Remember that we have the freedom to choose *any* surface S whose boundary coincides with C. An obvious choice would be to take S to be the unit disc on the xy-plane, but let us make our life more interesting by instead taking S to be the portion of the paraboloid $z = 1 - x^2 - y^2$ which lies above the xy-plane. The first thing to notice is that the intersection of this paraboloid with the xy-plane coincides with the unit circle C. Secondly, Stokes' Theorem tells us that

$$\oint_C \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{r} = \iint_S (\nabla\times\boldsymbol{F})\cdot\hat{\mathrm{n}}\,\mathrm{d}S.$$

We compute the curl of $\boldsymbol{F}$ as follows:

$$\nabla\times\boldsymbol{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{\mathrm{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = (1)\hat{\imath} - (-1)\hat{\jmath} + (1)\hat{\mathrm{k}} = \hat{\imath} + \hat{\jmath} + \hat{\mathrm{k}}.$$

To compute $\hat{\mathrm{n}}\,\mathrm{d}S$, observe that S is given by the graph of the function $f(x,y) = 1 - x^2 - y^2$, and so formula (48) applies, and implies that

$$\hat{\mathrm{n}}\,\mathrm{d}S = (-f_x, -f_y, 1)\,\mathrm{d}x\,\mathrm{d}y = (2x, 2y, 1)\,\mathrm{d}x\,\mathrm{d}y.$$

Carefully note that here we choose the unit normal vector with positive z-component, so that the orientation of S is compatible with that of C (in the sense explained above). It follows that

$$\iint_S (\nabla\times\boldsymbol{F})\cdot\hat{\mathrm{n}}\,\mathrm{d}S = \iint_S (1,1,1)\cdot(2x,2y,1)\,\mathrm{d}x\,\mathrm{d}y = \iint_U (2x+2y+1)\,\mathrm{d}x\,\mathrm{d}y,$$

where U denotes the unit disc on the xy-plane centred at the origin. The latter integral can be easily computed in polar coordinates (do it!). Alternatively, one may observe using symmetry that

$$\iint_U x \, \mathrm{d}x \, \mathrm{d}y = 0 = \iint_U y \, \mathrm{d}x \, \mathrm{d}y,$$

which in turn implies that

$$\iint_U (2x + 2y + 1) \, \mathrm{d}x \, \mathrm{d}y = \iint_U 1 \, \mathrm{d}x \, \mathrm{d}y = \mathrm{area}(U) = \pi.$$

This recovers the result from (60).

8.4. **Path independence.** Recall the notion of a simply-connected region in the plane from Definition 37. In three-dimensional space, we have the following definition.

Definition 64. *A region $D \subset \mathbb{R}^3$ is is* **simply-connected** *if every closed curve in D bounds a surface which is also contained in D.*

For example, note that the surface of a *sphere* is simply-connected but the surface of a *torus* is not simply-connected. See Figure 7 below. Another example comes from considering the whole space minus the z-axis removed, $\mathbb{R}^3 \setminus \{(0,0,z) : z \in \mathbb{R}\}$, which is *not* a simply-connected region. Observe that removing a single point is not enough, as the set $\mathbb{R}^3 \setminus \{(0,0,0)\}$ is simply-connected. This is in contrast with our earlier observation from §4.5 that removing a single point from the plane suffices, i.e. the region $\mathbb{R}^2 \setminus \{(0,0)\}$ is not simply-connected.

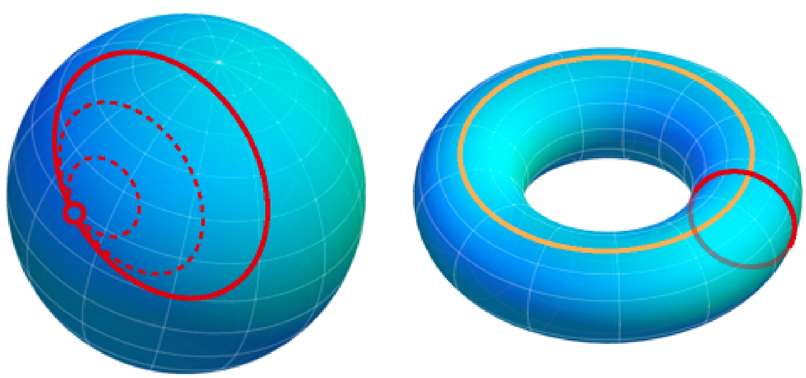

FIGURE 7. A sphere is simply-connected but a torus is *not* simply-connected.

We have already noticed the following result, which we can now state and prove in a rigorous way.

Theorem 65. *Let $\boldsymbol{F}$ be a vector field defined in a simply-connected region $D \subset \mathbb{R}^3$, such that $\nabla \times \boldsymbol{F} = \mathbf{0}$ on D. Then $\boldsymbol{F}$ is conservative.*

Proof. We have to check that

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = 0, \tag{61}$$

for any closed curve in D. Since D is simply-connected, we can find a surface S contained in D whose boundary coincides with the curve C. By Stokes' Theorem, we have that

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \iint_S (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}}\, \mathrm{d}S.$$

Since $\nabla \times \boldsymbol{F} = \mathbf{0}$ everywhere on D, and $S \subset D$, we have that $(\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}} = 0$ on S. Identity (61) follows, and the proof is complete. □

We have already seen that conservativeness implies path-independence, and we briefly recall the argument here. Let C be a closed curve, and let P_0, P_1 be distinct points on C. Write $C = C_1 - C_2$, where C_1 denotes a piece of C that connects P_0 to P_1, and C_2 denotes the complementary piece of C that also connects P_0 to P_1. In particular,

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \int_{C_1} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} - \int_{C_2} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r}.$$

If the left-hand side of this identity vanishes, then the two terms on the right-hand side are the same, and the result follows.

8.5. **Surface independence.** Consider a closed curve $C \subset \mathbb{R}^3$ and two distinct surfaces S_1 and S_2 whose boundaries are both equal to C. From Stokes' Theorem, it follows that

$$\iint_{S_1} (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}}\, \mathrm{d}S = \oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \iint_{S_2} (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}}\, \mathrm{d}S,$$

as long as the orientations of S_1, S_2 are compatible with that of C. In particular, this implies that

$$\iint_{S_1} (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}}\, \mathrm{d}S = \iint_{S_2} (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}}\, \mathrm{d}S, \tag{62}$$

and so the curl of a vector field is seen to have a "surface-independence property" that might seem strange at first. The answer comes, neatly enough, from Gauss' Divergence Theorem. To see why that is the case, consider the closed surface[20]

[20] The minus sign ensures that the unit normal vector $\hat{\mathrm{n}}$ to S is everywhere outward-pointing, which is the assumption of Gauss' Divergence Theorem.

$S = S_1 - S_2$, and denote by D the solid region bounded by S. Then we have that

$$\begin{aligned}\iint_{S_1} (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}}\, \mathrm{d}S - \iint_{S_2} (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}}\, \mathrm{d}S &= \iint_{S} (\nabla \times \boldsymbol{F}) \cdot \hat{\mathrm{n}}\, \mathrm{d}S \\ &= \iiint_D \operatorname{div}(\nabla \times \boldsymbol{F})\, \mathrm{d}V, \end{aligned} \tag{63}$$

where the first identity follows from the construction of the surface S, and the second identity follows from Gauss' Divergence Theorem. We claim that, for any vector field $\boldsymbol{F}$,

$$\operatorname{div}(\nabla \times \boldsymbol{F}) = 0.$$

Indeed, if $\boldsymbol{F} = (P, Q, R)$, then its curl is given by expression (56). In particular,

$$\begin{aligned}\operatorname{div}(\nabla \times \boldsymbol{F}) &= (R_y - Q_z)_x + (P_z - R_x)_y + (Q_x - P_y)_z \\ &= (R_{yx} - Q_{zx}) + (P_{zy} - R_{xy}) + (Q_{xz} - P_{yz}) = 0,\end{aligned} \tag{64}$$

because mixed partial derivatives commute. Going back to (63), this implies that

$$\iiint_D \operatorname{div}(\nabla \times \boldsymbol{F})\, \mathrm{d}V = \iiint_D 0\, \mathrm{d}V = 0,$$

and therefore (62) holds. This finishes the verification of the surface-independence property of the curl.

8.6. **Worked out example: Tetrahedron.** Consider the tetrahedron depicted in Figure 8, with vertices at

$$P_0 = (0,0,0),\ P_1 = (1,0,1),\ P_2 = (1,0,-1), \text{ and } P_3 = (1,1,0).$$

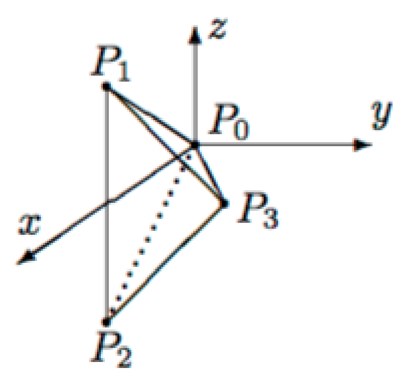

FIGURE 8. The tetrahedron in question. The faces $P_0P_1P_3$ and $P_1P_2P_3$ are visible, whereas the faces $P_0P_1P_2$ and $P_0P_2P_3$ are invisible.

(a) Say which orientation (order of vertices) of the boundary curve of each face is compatible with the choice of the normal vector pointing out of the tetrahedron.

(b) Compute the work done by the vector field $\boldsymbol{F} = yz\,\hat{\jmath} - y^2\,\hat{k}$ around the boundary curve of the face $P_0P_1P_3$ directly by using line integrals (using the orientation from part (a)).

(c) Use Stokes' Theorem to compute the work done around each of the four faces (including the one you computed directly in part (b)). Use the orientations from part (a).
(Note: The symmetry $z \mapsto -z$ exchanges two of the faces of the tetrahedron, and can be used to avoid one calculation. If you choose to use symmetry, you need to explain why it is legitimate.)

(d) The sum of the four values you found in part (c) should be zero. Explain this in two different ways:

(i) Geometrically, by considering the various line integrals that are being added together.

(ii) By using Gauss' Divergence Theorem to compute the flux of the curl of $\boldsymbol{F}$ out of the tetrahedron.

For part (a),

$$\begin{aligned}
&\text{For } P_0P_1P_2: \text{ from } P_0 \text{ to } P_2 \text{ to } P_1 \text{ back to } P_0;\\
&\text{For } P_0P_1P_3: \text{ from } P_0 \text{ to } P_1 \text{ to } P_3 \text{ back to } P_0;\\
&\text{For } P_0P_2P_3: \text{ from } P_0 \text{ to } P_3 \text{ to } P_2 \text{ back to } P_0;\\
&\text{For } P_1P_2P_3: \text{ from } P_1 \text{ to } P_2 \text{ to } P_3 \text{ back to } P_1.
\end{aligned}$$

For part (b), start by noting that the line segment from P_0 to P_1 can be parametrised as follows: $x = t, y = 0, z = t$ for $0 \le t \le 1$. It follows that

$$\int_{P_0P_1} yz\,\mathrm{d}y - y^2\,\mathrm{d}z = \int_0^1 0\,\mathrm{d}t = 0.$$

In a similar way, the line segment from P_1 to P_3 corresponds to $x = 1, y = t, z = 1-t$ for $0 \le t \le 1$, and so

$$\int_{P_1P_3} yz\,\mathrm{d}y - y^2\,\mathrm{d}z = \int_0^1 t(1-t)\,\mathrm{d}t - t^2(-\mathrm{d}t) = \int_0^1 t\,\mathrm{d}t = \frac{1}{2}.$$

Finally, the line segment from P_3 to P_0 is given by $x = t, y = t, z = 0$ for t going from 1 to 0, and so

$$\int_{P_3P_0} yz\,\mathrm{d}y - y^2\,\mathrm{d}z = \int_1^0 0\,\mathrm{d}t = 0.$$

Adding up these contributions, we see that the total work done by $\boldsymbol{F}$ around the boundary curve of the face $P_0P_1P_3$ equals $0+\frac{1}{2}+0=\frac{1}{2}$.

For part (c), start by noting that

$$\nabla\times\boldsymbol{F}=\begin{vmatrix}\hat{\imath} & \hat{\jmath} & \hat{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ 0 & yz & -y^2\end{vmatrix}=\left(\tfrac{\partial}{\partial y}(-y^2)-\tfrac{\partial}{\partial z}(yz)\right)\hat{\imath}=-3y\,\hat{\imath}.$$

By Stokes' Theorem, for each of the four faces S of the tetrahedron (with boundary curve C) we have that

$$\oint_C \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{r}=\iint_S(\nabla\times\boldsymbol{F})\cdot\hat{n}\,\mathrm{d}S=\iint_S(-3y\,\hat{\imath})\cdot\hat{n}\,\mathrm{d}S.$$

Hence we have to find the flux of the vector field $\boldsymbol{G}=\nabla\times\boldsymbol{F}=-3y\,\hat{\imath}$ through each face.

Since the face $P_0P_1P_2$ is contained in the xz-plane (corresponding to $y=0$), the outward unit normal is given by $\hat{n}=-\hat{\jmath}$. In this case, we have that $\boldsymbol{G}\cdot\hat{n}=0$, and so the flux is zero.

The face $P_1P_2P_3$ is contained in the plane $x=1$, and so the outward unit normal is $\hat{n}=\hat{\imath}$, and $\boldsymbol{G}\cdot\hat{n}=(-3y,0,0)\cdot(1,0,0)=-3y$. We compute:

$$\begin{aligned}\iint_{P_1P_2P_3}\boldsymbol{G}\cdot\hat{n}\,\mathrm{d}S&=\iint_{P_1P_2P_3}-3y\,\mathrm{d}S=\int_0^1\int_{-(1-y)}^{1-y}-3y\,\mathrm{d}z\,\mathrm{d}y\\&=\int_0^1-6y(1-y)\,\mathrm{d}y=\left[-3y^2+2y^3\right]_0^1=-1.\end{aligned}$$

We now come to the face $P_0P_1P_3$. A normal vector (pointing outwards) to this face is given by

$$\boldsymbol{N}=\overrightarrow{P_0P_1}\times\overrightarrow{P_0P_3}=(1,0,1)\times(1,1,0)=(-1,1,1).$$

In particular, the face $P_0P_1P_3$ is contained in the plane $-x+y+z=0$, i.e. the graph $z=x-y$ of $f(x,y)=x-y$. So

$$\hat{n}\,\mathrm{d}S=(-f_x,-f_y,1)\,\mathrm{d}x\,\mathrm{d}y=(-1,1,1)\,\mathrm{d}x\,\mathrm{d}y,$$

and consequently $\boldsymbol{G}\cdot\hat{n}\,\mathrm{d}S=(-3y,0,0)\cdot(-1,1,1)\,\mathrm{d}x\,\mathrm{d}y=3y\,\mathrm{d}x\,\mathrm{d}y$. The projection of the face $P_0P_1P_3$ on the xy-plane is a triangle with vertices at $(0,0)$, $(1,0)$ and $(1,1)$, and so

$$\iint_{P_0P_1P_3}\boldsymbol{G}\cdot\hat{n}\,\mathrm{d}S=\int_0^1\int_0^x 3y\,\mathrm{d}y\,\mathrm{d}x=\int_0^1\tfrac{3}{2}x^2\,\mathrm{d}x=\left[\tfrac{x^3}{2}\right]_0^1=\frac{1}{2}.$$

The symmetry $(x,y,z)\to(x,y,-z)$ exchanges the two faces $P_0P_1P_3$ and $P_0P_2P_3$, so the two normal vectors are symmetric to each other (the orientations match). Since

$\boldsymbol{G} = -3y\,\hat{\imath}$ is also preserved by this symmetry, the flux through $P_0P_2P_3$ is the same as through $P_0P_1P_3$, namely $\frac{1}{2}$.
Alternatively, we can notice that $P_0P_2P_3$ is contained in the plane $z = -x + y$, so $\hat{n}\,\mathrm{d}S = -(1, -1, 1)\,\mathrm{d}x\,\mathrm{d}y$ (the negative sign is there so that $\hat{n}$ points downwards) and $\boldsymbol{G} \cdot \hat{n}\,\mathrm{d}S = 3y\,\mathrm{d}x\,\mathrm{d}y$. The projection of the face $P_0P_2P_3$ onto the xy-plane is again the triangle with vertices $(0, 0)$, $(1, 0)$ and $(1, 1)$, and the calculation proceeds as previously to yield $\frac{1}{2}$.

We finally come to part (d).
For (i), when adding together the four answers from (c), we compute the work along a curve which passes *twice* over each of the six edges of the tetrahedron. However each edge is transversed once with each orientation, and so the various contributions cancel each other out. For example, the edge P_0P_1 is encountered once for the face $P_0P_1P_2$ (it is then oriented from P_1 to P_0) and once for the face $P_0P_1P_3$ (it is then oriented from P_0 to P_1); the sum of the two contributions is zero.
For (ii), notice that Stokes' Theorem implies

$$\oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \iint (\nabla \times \boldsymbol{F}) \cdot \hat{n}\,\mathrm{d}S.$$

for each face, and so the sum of the four line integrals equals the flux of $\nabla \times \boldsymbol{F} = -3y\,\hat{\imath}$ out of the tetrahedron. By Gauss' Divergence Theorem, we have that

$$\iint_S (\nabla \times \boldsymbol{F}) \cdot \hat{n}\,\mathrm{d}S = \iiint_D \operatorname{div}(\nabla \times \boldsymbol{F})\,\mathrm{d}V,$$

where S denotes the tetrahedron and D the solid region encapsulated by S. But, as we have already seen in (64), $\operatorname{div}(\nabla \times \boldsymbol{F}) = 0$, and so the total flux is zero.